

Data Science Project Lifecycle: Project A

Group 1

Mustafiz Bakhsh (MB) -w2066721

Rayhanur Rahman (RR) - w2073057

Sameer Janal (SJ) - w2117206

Muhtasim Choudhury (MC) - w2068659

Dominykas Revinas (DR) - w1987563

Executive Summary	2
Background and Problem Domain	2
Project Objectives	4
Preliminary Analysis RR (W2073057) (W1987563) DR, SJ (W2117206).....	4
CATWOE Analysis (MB W2066721)	5
Interview (Everyone)	7
Notes from Interview (RR W2073057)	8
Proposed Data Sources (RR w2073057)	10
Initial Thoughts	11
Rich Picture (RR W2073057).....	11
Project Plan	12
Meeting Schedule (MB W2066721)	13
Roles and Responsibilities (MB W2066721).....	13
Work Breakdown Structure (WBS) (MB W2066721)	15
Gant Chart (MB W2066721)	18
Requirements and Expected Deliverables (MB W2066721)	19
Risk Analysis.....	20
Dataset Overview (RR W2073057)	21
Dataset 1 – 20260213 Fleet Performance	22
Dataset 2 – Fleet Delay Data from P2301 to P2611	23
The Fleet Delay dataset contains 3,925 rows and 55 columns. Each row represents a delay incident affecting train services. Compared to the Fleet Performance dataset, this dataset focuses more directly on delay severity and operational impacts.....	24
Dataset 3 – Fleet Planning Log.....	24
How the Datasets Work Together	25
Summary.....	25
Fleet Delay Dataset.....	26
Fleet Performance Dataset	26
Fleet Planning Log Dataset	26
Common Join Key Between Datasets.....	26
Cleaning Datasets.....	26
Cleaning Delay dataset	27

Cleaning Performance dataset:.....	27
Cleaning Planning dataset	28
Reasons for Not Merging the Datasets	29
Exploratory Data Analysis	29
Figure 1: Fleet Delay Incidents by Period (P2201 – P2611) SJ(W2117206)	30
Figure 2: Fleet Delay (minutes) by Operational Period (P2201 – P2611) SJ(W2117206)	30
Figure 3: Operational Issues by Late Started Events (Depot) SJ(W2117206).....	31
Figure 4: Distribution of Delay Minutes (RR W2073057)	32
Figure 5: Top 10 Delay Causes (RR w2073057).....	33
Figure 6: Top 10 Locations by Total Delay Minutes (RR w2073057).....	34
Figure 7: Trains Affected Vs Delay Minutes (RR w2073057)	36
Figure 8: Root Cause Severity (RR w2073057)	37
Correlation Analysis (RR w2073057)	38
Exploratory Data Analysis Conclusion	38
Data Preparation, Modelling, and Evaluation (RR w2073057)	39
Distribution of Operational Pressure (RR w2073057).....	40
Feature Engineering (RR w2073057).....	40
Assumptions and Modelling Decisions (RR w2073057).....	41
One – Hot Encoding (RR w2073057).....	41
Predictive Modelling (RR w2073057)	42
Decision Tree Model (RR w2073057).....	42
Random Forest Model (RR w2073057)	43
Overfitting Check (RR w2073057)	44
Model Comparison (RR w2073057).....	45
Modelling Justification and Alternatives (RR w2073057)	46
Limitations of the Model (RR w2073057)	46
Forward-Looking Modelling Extension (MB W2066721)	47
Model Evaluation (MB W2066721)	47
Operational Application During Disruption	48
Forecast Horizon for Fleet Availability Risk	48
Recommendations (MB W2066721).....	49

What are the main operational pain points caused by limited spare rolling stock? (MB W2066721)	50
Which types of disruption cause the greatest systemic impact? (MB W2066721)	50
How is fleet availability currently assessed, and what are the limitations of existing processes? (MB W2066721).....	50
Which visual indicators best represent availability risk? (MB W2066721)	51
How could the developed model be utilised during disruption scenarios? (MB W2066721)	51
Conclusion (MB W2066721).....	51
Github link:	52
Referencing	53
Individual Reflections	55
Rayhanur Rahman (RR w2073057).....	55
Mustafiz Bakhsh (MB W2066721)	57
Sameer Janal (SJ) W2117206.....	63

Executive Summary

(W1987563) DR

Reliability in the railway industry plays a critical role in the performance of the railway services, especially for the passengers using the services. The delays in the railway services increase the duration of the journey, hence impacting the efficiency of the services offered. The London North Eastern Railway (LNER) services operate on the East Coast Main Line, connecting important towns such as London, York, Leeds, Newcastle, and Edinburgh.

In this project, our team uses three datasets to analyse the availability of the fleets in the railway services as well as the delays in the services. The datasets used in the project include the Fleet Performance, Fleet Delay, and Fleet Planning Log datasets, which offer critical information on the delays, fleet performance, as well as the fleet planning, hence facilitating the understanding of the relationship between the delays, fleet performance, and the fleet planning.

Background and Problem Domain

(W1987563) DR

Reliability in railway transport is one of the key determinants in delivering quality passenger transportation services in line with customer needs. Railway service delays reduce efficiency and passenger satisfaction levels while also having a negative effect on overall railway network operational performance. This explains why railway reliability is an issue of great concern to the regulators and operators due to its importance in maintaining quality service delivery and performance. (Office of Rail and Road, 2025)

East Coast Main Line (ECML) is an important railway line managed by London North Eastern Railway (LNER). It links some of the busiest cities in UK such as London, York, Leeds, Newcastle, and Edinburgh. In this case, the high passenger demands and tight scheduling require a certain level of precision in operation and punctuality. Any delay in the services on ECML can lead to more delays in other trains because of the limited capacity of train services on this route (Network Rail, 2024).

Railways operations occur in complex, closely-coupled operational environments where any disturbance in the system can have a knock-on effect in other systems (Goverde, 2010). Therefore, understanding the cause of railway delays is critical in improving railway operational performance. Studies indicate that railway disruptions can result from a variety of problems, including infrastructural, rolling stock failure, signalling problems, and operating environment (Carey and Carville, 2003). In this regard, it becomes essential to understand these variables and the role they play in causing railway disruptions.

Recently, there has been increased interest in the use of data science and machine learning in addressing railway performance issues. Data science approaches allow railway operators to make observations using vast amounts of railway data, which may not be easy to observe using traditional methods (Corman and Kecman, 2018). In this respect, it is possible to predict indicators of disruptions and high operational pressure in railway network using data analysis techniques.

Three datasets were used to analyse the railway delay pattern in the given project. The first one is the historic operational incidents dataset that contains data regarding operational incidents. It includes such information as engineering faults and operational service disruptions. The second dataset is delay information and covers delay minutes, delay cause, and the number of trains involved in incidents. The delay minutes are categorized in terms of infrastructures, rolling stock, and signalling issues and help to identify the reasons behind the disruptions (Office of Rail and Road, 2025). The last dataset concerns fleet planning logs and provides information about planning periods, date of reports, and train units. This dataset helps to obtain an understanding of fleet availability planning capacity.

The combination of all of the described datasets will help to establish connections between operational disruptions, delay duration and their reasons, and fleet planning limitations. Knowledge of the interrelation between the factors under consideration might be useful to understand the situations that lead to increased operational stress and result in low fleet availability. However, no connections have been established yet, thus, more work on exploring and predicting the situation must be done. Considering that the availability of operational data increases significantly at railways, the possibility to use data science and data analysis should be explored. Exploratory analysis will allow obtaining signals indicating increased operational stress and help in developing proactive strategies of fleet management. For this reason, the present project focuses on investigating the application of data science methods to LNER disruptions and fleet availability.

References

Carey, M. and Carville, S. (2003) 'Scheduling and platforming trains at busy complex stations', *Transportation Research Part A*, 37(3), pp.195–224.

Corman, F. and Kecman, P. (2018) 'Stochastic prediction of train delays in real-time railway traffic management', *Transportation Research Part B*, 105, pp.472–488.

Goverde, R. (2010) 'A delay propagation algorithm for large-scale railway traffic networks', *Transportation Research Part C*, 18(3), pp.269–287.

Network Rail (2024) *East Coast Main Line Performance and Reliability Report*. Available at: <https://www.networkrail.co.uk>

Office of Rail and Road (2025) *Passenger Rail Performance Statistics*. Available at: <https://dataportal.orr.gov.uk>

Project Objectives

The following objectives have been determined according to the project brief and stakeholder requirements in priority order.

- 1) **Pinpoint the key risk components associated with fleet availability risk** - Find out which operational, maintenance and environmental factors have the most impact on East Coast Main Line network fleet availability, owned by LNER.
- 2) **Create a forecasting framework that will assist in predicting high operational pressure periods** - Create and test a candidate machine learning model(s) that identify patterns in historical delay and performance data that have previously led to fleet availability risk.
- 3) **Research operational pressure at depot level** – We will obtain a better understanding of which depots incur the most significant frequency and magnitude of delay incidents to aid maintenance and allocation decisions.

- 4) **Evaluate dataset quality and suitability** – Assess the completeness and reliability of the Fleet performance, Fleet Planning and Fleet Delay datasets and establish a data preparation pipeline to enable ongoing analysis.
- 5) **Translate model outputs into actionable recommendations** - Offer signals and mitigation recommended strategies operationally relevant to the Engineering and Control Function of LNER. These will be integrated within a 24–48-hour forecasting horizon.

Preliminary Analysis RR (W2073057) (W1987563) DR, SJ (W2117206)

LNER operates a tightly optimised railway network connecting major UK cities. Following a recent timetable expansion of approximately 25%, the system now runs with limited spare rolling stock.

Fleet availability risk refers to the possibility that an insufficient number of operational train sets are available to meet timetable requirements. When availability falls below required levels, the consequences can include service cancellations, knock-on delays, and increased operational pressure on control and engineering teams.

From our initial review of the project brief and supporting materials, it is clear that fleet availability is not simply an engineering issue. It is a broader operational risk that affects service reliability, financial performance, and customer satisfaction.

Several operational datasets have been used to explore the factors that influence fleet availability. The Fleet Delay dataset (P2201 – P2611) was used to analyse the frequency of incident delays across operational periods. Using this dataset, insights on how frequently disruptions occur over time.

Having used this dataset, the Fleet Performance dataset was also used to investigate the delay minutes across the same operational periods. The purpose of the Fleet Performance dataset was to assess the severity of these disruptions.

From these two datasets, the initial thought was that while there are consistent delay incidents occurring across each operational period, certain periods showed spikes in delay minutes. This suggests that there were high counts of disruption.

Logs from the Late Starts sheet, derived from the Fleet Planning dataset were used to evaluate the depot- level operational issues. When analysing the Late Starts sheet, it was

revealed that there were depots with a higher frequency of late start events. This connotes that some depots potentially face increased operational pressure.

In conclusion, three key aspects of fleet availability risk were emphasised:

- Delay frequency
- Delay severity
- Depot location where these delays took place

CATWOE Analysis (MB W2066721)

Letter	Element	Description
C	Client	Primary clients are the Engineering and Control teams that operate LNER. Passengers are the secondary clients of the company. Engineering teams require this information to schedule maintenance of the fleet. Control teams require this information to make operational decisions. Passengers benefit from improved availability as fewer services are cancelled due to a lack of rolling stock
A	Actor	The actors are those responsible for carrying out the transformation process: • Group 1 Data Science team — developing the predictive model and analytical framework

		• LNER Engineering staff — providing domain knowledge and validating model outputs • LNER Control room staff — end users of the early warning system • University of Westminster module team — overseeing the project and providing feedback • LNER client representative — providing datasets and formative feedback on progress
T	Transformation	Transformation from managing fleet issues as they occur to predicting when fleets will be available for customers. Input: Data on the fleets that is currently managed manually with no early warning of potential fleet issues. Output: A model that predicts the availability of the company's fleets by period, depot, and fleet class to provide management with an early warning of potential fleet issues.
W	Worldview	Underlying worldview is that the risk of rolling stock unavailability is preventable if the appropriate tools and data are deployed. With minimal spare rolling stock and with 25% more trains to operate on the LNER timetable, any failure in LNER fleets incurs costs of over £10,000 per cancelled train. The project intends to use data science methods to drastically reduce the costs and reputational damage that the LNER currently incurs.
O	Owner	The system owner is LNER as the client organisation commissioning this analytical solution. LNER retains ownership of all operational data provided and will be the ultimate decision-maker regarding whether and how the model outputs are adopted. Within the project, Group 1 (led by Project Manager Mustafiz Bakhsh, MB w2066721) is responsible for delivering the solution within the constraints of the module brief. The module leader, Dr Philip Worrall, holds academic oversight of all project deliverables.
E	Environment	The following constraints will define the environment in which the project will take place:

		• Data limitations – there is no available engineering or fault data for LNER; instead, data on the delays, cancellations and planning of the LNER train services will be used • Time constraint – the project is to be completed within an 8 week period with deliverables due 19 March, 27 March and 7 April 2026 • Technical constraints – only the software packages and tools available to the team will be used within the report, including Python, Google Collab, Power BI and Excel • Regulatory context – the LNER is regulated by the Office of Rail and Road, from which the performance data for the company is published • Organisational constraints – the LNER already operates its timetable to optimise efficiency in its current operations, which limits the flexibility of the organisation in responding to availability warnings for its trains
--	--	--

Interview (Everyone)

1. In day-to-day operations, which fleet availability issues create the most pressure for control or engineering teams?
2. Based on your experience, which disruption types tend to have the biggest knock-on effects across the network?
3. Are there particular depots, fleet classes, or time-of-day periods where availability risk is consistently higher?
4. How is fleet availability currently monitored or assessed in practice, and where do existing processes fall short in providing early warning?
5. When spare rolling stock (extra trains not currently in use) is limited, how are unexpected faults prioritised operationally?
6. At what point would an early-warning signal still be actionable for teams?
7. How far ahead would a fleet availability forecast be most useful in practice — hours, days, or longer?

Notes from Interview (RR W2073057)

Key Themes

- The company is moving towards better use of data, but many processes are still manual.
- The fleet system is tightly optimised, with limited spare trains available.
- Fleet availability management is currently mostly reactive rather than predictive.

- Several factors affect fleet availability, including component wear, suppliers, depots, and weather.
- Train cancellations have significant financial and reputational impact.

1. Company Context

Digital operations team established around 5 years ago

Data Science team is around 3 years old

- 200–300+ systems generating operational data
- Historically slow data access
- Company is now building centralised data infrastructure

2. Fleet Context

- Around 250 total trains
- Around 150 trains required operational daily
- Limited spare rolling stock
- 25% timetable capacity increase

System now tightly optimised

3. How Fleet Availability is Currently Managed

Largely reactive approach

- Heavy manual maintenance checks
- Sensor data exists but is underused
- Engineering team not strongly data-driven

No integrated predictive system

4. Causes of Availability Risk

Wheel / Traction Wear:

- High wear components
- Can degrade gradually
- May not be picked up early

Supplier Differences:

- Some fleet classes better supported than others

- Response speed varies

Maintenance Schedules:

- Regular checks exist

Unpredicted failures still occur

In-Service Failures:

- Failures can occur during operation
- Trains may be removed mid-service
- Can create cascading operational impact

Depot Constraints

Main depots mentioned:

- London, York, Edinburgh

Weather and Track Interaction

Examples mentioned:

- Snow
- Flooding
- Animals on track

5. Financial Impact

Train cancellation cost around £10,000+ per train

- Reputation damage

Performance penalties

6. System Characteristic

Tightly coupled network

- Limited flexibility during peak times
- Regulatory constraints limit reshuffling
- Some days allow flexibility, others do not

Through the interview, for us as a group it provided us with useful insight into the London Northeastern Railway and the operational challenges that come along with it. Fleet availability

management is currently largely reactive; this is mostly because of the heavy reliance on manual maintenance checks and limited use of available sensor data. The interview gave us an insight on that they have a very limited spare capacity if things do go wrong, meaning that even small failures can quickly lead to cancellations and wider disruption. Several risk factors were mentioned, including part wear, supplier response times, depot constraints, and environmental factors such as weather or track obstructions. From these insights such things as operational data and predictive approaches could help them find the risks a lot earlier and help them with fleet availability.

Proposed Data Sources (RR w2073057)

ORR Table 3103 – Quarterly Performance by Operator:

This dataset contains LNER's quarterly performance data, including Cancellations and Significant Lateness (CaSL) and trains planned. CaSL may act as an indicator of worsening service performance and potential availability pressure. Analysing trends over time will help identify periods where performance declines.

<https://dataportal.orr.gov.uk/statistics/performance/passenger-rail-performance/table-3103-historic-passenger-trains-planned-ppm-and-casl-by-operator/>

ORR Table 3184 – Delay Minutes by Operator and Cause:

This dataset breaks down delay minutes by cause category. It allows us to examine what types of issues contribute most to disruption and whether certain causes are linked to performance deterioration. <https://dataportal.orr.gov.uk/statistics/performance/passenger-rail-performance/table-3184-delay-minutes-by-operator-and-cause-periodic/>

Weather Data (Station-Level Historical Data):

Historical weather data, such as temperature and rainfall, may be included as an external factor. Weather conditions can influence operational stability and disruption levels.

<https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=UKE00107650&startDate=2018-01-01&endDate=2025-12-31&dataTypes=TMAX,TMIN,PRCP,SNOW,WND&format=csv>

Although direct engineering data is not currently available, combining performance data with delay causes and weather provides a practical starting point for analysing availability risk. Further exploratory analysis will be conducted in the next stage.

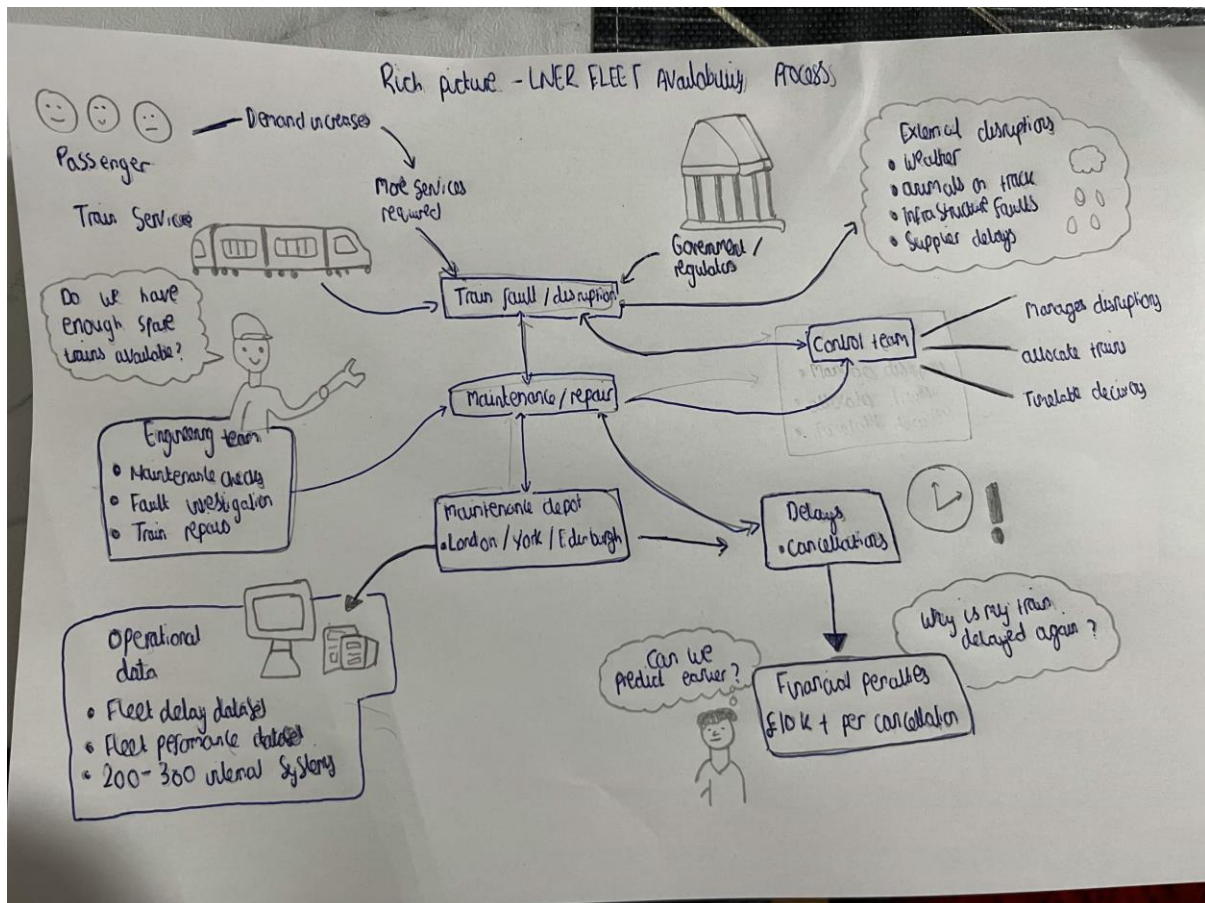
At this stage, we are considering modelling availability risk using performance indicators such as CaSL, either by analysing changes over time or identifying periods of significant performance decline. The final modelling approach will be refined after exploratory analysis.

Initial Thoughts

For the initial approach to the problem, we believed that analysing the available data on factors like delays and cancellations would be sufficient to determine any patterns with relation to the availability of the aircraft. Through further discussion on the problem, though, we became aware that there are other factors related to availability that should also be considered.

One major challenge is the absence of direct engineering or fault datasets within the available data. This means maintenance effectiveness and fault behaviour cannot be observed directly. Instead, we must rely on operational performance measures, such as delays and cancellations, to give us an indication of how maintenance and fault issues might be affecting fleet availability.

Rich Picture (RR W2073057)



The diagram highlights the role of the control team, who are responsible for managing disruptions, allocating trains, and making timetable decisions in response to incidents. External factors such as weather conditions, animals on the track, infrastructure faults, and supplier delays are shown as additional sources of disruption that are outside direct operational control.

Operational data plays an important role in this system, as it is collected from multiple internal systems and used to monitor fleet performance and delays. However, the process is largely reactive, meaning actions are taken after disruptions occur rather than being predicted in advance.

The outcomes of these disruptions include delays and cancellations, which lead to passenger dissatisfaction and financial penalties, estimated at over £10,000 per cancellation. The inclusion of questions such as “Can we predict earlier?” highlights the key motivation of the project, which

is to move from a reactive approach to a more proactive, data-driven system that can forecast high-risk situations before they occur.

Project Plan

Project Manager: Mustafiz Bakhsh

Communication – Via Teams

Meeting Schedule (MB W2066721)

Date	Duration	Type	Attendance	Topic discussed
10/02/26	45 minutes	In- person	Everyone	• Understand coursework brief and marking criteria • Researched LNER background and operational context • Classified overall project direction
17/02/2026	30 minutes	Online	Everyone	• Allocated individual responsibilities • Discussed key themes for client interview • Drafted and refined priority questions to ask • Agreed on priority questions to ask
25/02/2026	1 hour	Online	Everyone	• Reviewed structure of initial draft • Identified key topics for preliminary analysis • Discussed available datasets • Discussed potential data sources and modelling direction
07/03/2026	45 minutes	Online	Everyone	• Reviewed project progress and current phase • Allocated tasks • Confirmed datasets (Fleet Delay, Fleet performance, Fleet planning log) • Set up GitHub repository • Discussed modelling approach and variable selection

12/03/2026	20 minutes	Online	Everyone	• Raised issue initially merging data sets, data sets don't align well for full merge • Writing the Data Quality and Preparation section including row counts before and after cleaning join methodology • Compile all sections into a document with correct layout
19/03/2026	30 minutes	Online	Everyone	• Started drafting and reviewing model evaluations • Create a table of contents page in the report • Brainstorm and discuss ideas for upcoming presentation
25/03/2026	20 minutes	Online	Everyone	• From the data modelling conducted we discuss how to draft and finalise the final recommendations for the client • Begin creating PowerPoint slides for the upcoming presentation • Ensuring all references and github link are present at the end of the report
01/04/2026	20 minutes	Online	Everyone	• Improving and making changes to final report based on feedback given from our group • Discuss everyone's own contribution to start on their individual feedback

Roles and Responsibilities (MB W2066721)

Responsibilities	Team Member
Background research and problem domain	Everyone
Preliminary analysis	MB w2066721, RR w2073057
Initial thoughts and project direction	Everyone

Project plan, WBS, Gantt chart, Rich picture and CATWOE	MB w2066721, SJ w2117206, DR w1987563, RR w2073057
Interview questions	Everyone
Interview notes and key themes	MB w2066721, DR w1987563, RR w2073057
Analysis of risks	DR w1987563, MB w2066721
Problem structuring using diagrams	MB w2066721
Collect and upload datasets	RR w2073057, MC w2068659
Data preparation, cleaning and integration	MC w2068659, RR w2073057
Exploratory data analysis (EDA)	SJ w2117206, RR w2073057
Feature engineering and modelling	RR w2073057
Model evaluation and forecasting	RR w2073057, MB w2066721
Trends analysis and recommendations	Everyone
Report write-up and proofreading	DR w1987563, MB w2066721
GitHub repository	RR w2073057
Presentation to client	Everyone
Individual reflection	Everyone

Work Breakdown Structure (WBS) (MB W2066721)

ID	Phase / Task	Subtask Description	Owner	Duration	Depends On
1	Initial Project Analysis		All		-
1.1		Understand project brief and scope	All	1 day	-
1.2		Define project objectives and deliverables	Mustafiz	1 day	1.1
1.3		Conduct background research on LNER	Dominyckas	2 days	1.1
1.4		Prepare and conduct client interview	All	1 day	1.1
1.5		Write preliminary analysis and initial thoughts	Sameer/ Rayhanur	2 days	1.4
1.6		Compile and submit initial draft report	Mustafiz	1 day	1.5
2	Project Planning		Mustafiz		1
2.1		Create Work Breakdown Structure (WBS)	Mustafiz	1 day	1.2
2.2		Build Gantt chart covering all 4 weeks	Mustafiz	1 day	2.1
2.3		Create Activity Network Diagram	Mustafiz	1 day	2.1
2.4		Write CATWOE analysis	Mustafiz	1 day	1.4
2.5		Define team roles and responsibilities	Mustafiz	1 day	1.2
2.6		Write risk analysis (Data/Analytical/Operational/External)	Dominyckas	2 days	2.1
2.7		Set up Google Drive folder structure	Rayhanur	1 day	-
2.8		Set up GitHub repository	Rayhanur	1 day	-
2.9		Maintain meeting logs throughout project	Mustafiz	Ongoing	-
3	Data Preparation and Integration		Muhtasim		2
3.1		Upload all LNER datasets to Drive and GitHub	Rayhanur	1 day	2.7

3.2		Load and inspect Fleet Delay dataset (P2301-P2611)	Muhtasim / Rayhanur	1 day	3.1
3.3		Load and inspect Fleet Performance dataset	Muhtasim / Rayhanur	1 day	3.1
3.4		Load and inspect Fleet Planning Log and Depot Sheets	Muhtasim / Rayhanur	1 day	3.1
3.5		Evaluate data quality for each dataset	Muhtasim	1 day	3.2
3.6		Clean Fleet Delay dataset	Muhtasim	1 day	3.5
3.7		Clean Fleet Performance dataset	Muhtasim	1 day	3.5
3.8		Identify common join key across datasets	Muhtasim	1 day	3.4
3.9		Merge all datasets into master dataframe	Muhtasim	1 day	3.6
3.10		Write Dataset Acquisition section	Dominyckas	1 day	3.1
3.11		Write Data Preparation section in report	Dominyckas	2 days	3.9
4	Exploratory Data Analysis (EDA)		Sameer		3
4.1		EDA: Bar chart of delay causes by frequency	Sameer	1 day	3.6
4.2		EDA: Time-series of fleet performance across periods	Sameer	1 day	3.6
4.3		EDA: Depot-level breakdown of delays and risk	Sameer	1 day	3.6
4.4		EDA: Correlation heatmap of key variables	Rayhanur	1 day	3.9
4.5		Feature engineering: create availability risk proxy variable	Rayhanur	2 days	3.9
4.6		Create Power BI / Python dashboard	Sameer	3 days	4.3
4.7		Write EDA section in report	Sameer	2 days	4.3
5	Modelling and Implementation		Rayhanur		4

5.1		Justify modelling approach (vs alternatives in brief)	Rayhanur	1 day	4.4
5.2		Build baseline model (Logistic Regression / Decision Tree)	Rayhanur	2 days	4.5
5.3		Build primary model (Random Forest / XGBoost)	Rayhanur	2 days	5.2
5.4		Evaluate model performance (accuracy, F1, R ²)	Rayhanur	1 day	5.3
5.5		Feature importance analysis	Rayhanur	1 day	5.3
5.6		Forecast availability risk for future periods	Rayhanur	2 days	5.4
5.7		Write Modelling section in report	Mustafiz/Rayhanur	2 days	5.4
6	Draft Project Report for Client		Mustafiz		5
6.1		Compile all sections into second draft	Mustafiz	1 day	5.7
6.2		Team review — all members check and flag issues	All	2 days	6.1
6.3		Address all feedback and submit draft	Mustafiz	1 day	6.2
7	Live Presentation		All		6
7.1		Create presentation slides (5-7 slides)	Rayhanur	2 days	6.3
7.2		Rehearse presentation	All	1 day	7.1
7.3		Deliver live presentation to client	All	1 day	7.2
8	Final Project Report		All		7
8.1		Address client and lecturer feedback from draft and presentation	Mustafiz	2 days	7.3
8.2		Write Key Insights and Recommendations section	Mustafiz	2 days	5.6
8.3		Write Assumptions and Limitations section	Rayhanur	1 day	5.4
8.4		Individual reflection — Mustafiz (MB)	Mustafiz	2 days	-
8.5		Individual reflection — Rayhanur (RR)	Rayhanur	2 days	-

8.6		Individual reflection — Muhtasim (MC)	Muhtasim	2 days	-
8.7		Individual reflection — Sameer (SJ)	Sameer	2 days	-
8.8		Individual reflection — Dominyckas (DR)	Dominyckas	2 days	-
8.9		Final GitHub audit — all code runs cleanly	Rayhanur	1 day	5.6
8.10		Final proofread — split pages between all members	All	1 day	8.3
8.11		Compile and submit final report	Mustafiz	1 day	8.10

Gant Chart (MB W2066721)

ID	Task / Subtask	Owner	Week 1 7–13 Mar	Week 2 14–20 Mar	Week 3 21–27 Mar	Week 4 28 Mar–3 Apr	Week 5 4–7 Apr	Milestone / Note
2.2	Build Gantt chart	Mustafiz						
2.3	Create Activity Network Diagram	Mustafiz						
2.4	Write CATWOE analysis	Mustafiz						
2.5	Define team roles and responsibilities	Mustafiz						
2.6	Write risk analysis table	Dominyckas						
2.7	Set up Google Drive and GitHub	Rayhanur						
2.8	Maintain meeting logs	Mustafiz						
3	PHASE 3: Data Preparation							
3.1	Upload datasets to Drive and GitHub	Rayhanur						
3.2	Load and inspect Fleet Delay dataset	Wafi						
3.3	Load and inspect Fleet Performance dataset	Wafi						
3.4	Load Planning Log and Depot Sheets	Wafi						
3.5	Evaluate data quality for each dataset	Wafi						
3.6	Clean Fleet Delay dataset	Wafi						
3.7	Clean Fleet Performance dataset	Wafi						
3.8	Identify common join key across datasets	Wafi						
3.9	Merge datasets into master dataframe	Wafi						
3.10	Write Dataset Acquisition section	Dominyckas						
3.11	Write Data Preparation section	Dominyckas						
4	PHASE 4: Exploratory Data Analysis (EDA)							
4.1	EDA: Bar chart of delay causes	Sameer						
4.2	EDA: Time-series of fleet performance	Sameer						
4.3	EDA: Depot-level breakdown	Sameer						
4.4	EDA: Correlation heatmap	Rayhanur						
4.5	Feature engineering: risk proxy variable	Rayhanur						
4.6	Create Power BI / Python dashboard	Sameer						
4.7	Write EDA section in report	Sameer						
5	PHASE 5: Modelling and Implementation							
5.1	Justify modelling approach	Rayhanur						
5.2	Build baseline model	Rayhanur						
5.3	Build primary model (RF / XGBoost)	Rayhanur						
5.4	Evaluate model performance	Rayhanur						
5.5	Feature importance analysis	Rayhanur						
5.6	Forecast availability risk	Rayhanur						
5.7	Write Modelling section in report	Mustafiz						

6	PHASE 6: Draft Report for Client							
6.1	Compile all sections into second draft	Mustafiz						
6.2	Team review and feedback	All						
6.3	Address feedback and submit draft	Mustafiz						DEADLINE: 19 Mar
7	PHASE 7: Live Presentation							
7.1	Create presentation slides (5-7 slides)	All						
7.2	Rehearse presentation	All						
7.3	Deliver live presentation to client	All						DEADLINE: 27 Mar
8	PHASE 8: Final Project Report							
8.1	Address feedback from draft + presentation	Mustafiz						
8.2	Write Key Insights and Recommendations	All						
8.3	Write Assumptions and Limitations	Rayhanur						
8.4	Individual reflections — all members	All						
8.5	Final GitHub audit	Rayhanur						
8.6	Final proofread — all members	All						
8.7	Compile and submit final report	Mustafiz						DEADLINE: 7 Apr

Requirements and Expected Deliverables (MB W2066721)

The four deliverables each relate to a specific topic and phase of the project lifecycle.

The first deliverable was the Initial Draft Project Analysis. On 26th February 2026, a brief report was submitted outlining the preliminary analysis, initial thoughts, and project plan. The module team provided the group with formative feedback through this submission.

The second deliverable was the Draft Project Report for the client. An extended report was submitted on 19th March 2026, outlining the group's progress to date. The report articulated the analytical approach and methodology undertaken. The purpose of this submission was to receive formative feedback from LNER.

The third deliverable was the Live Presentation to the client, delivered on 27th March 2026. The presentation was ten minutes in duration and summarised the work carried out to date. This was a formative submission, with verbal feedback provided by the industry client.

The fourth and final deliverable is the Final Project Report, due 21st April 2026. All technical code and implementation files will be hosted on the group's GitHub repository. This represents the complete analytical solution developed in response to the LNER project brief and is the summative submission for this module.

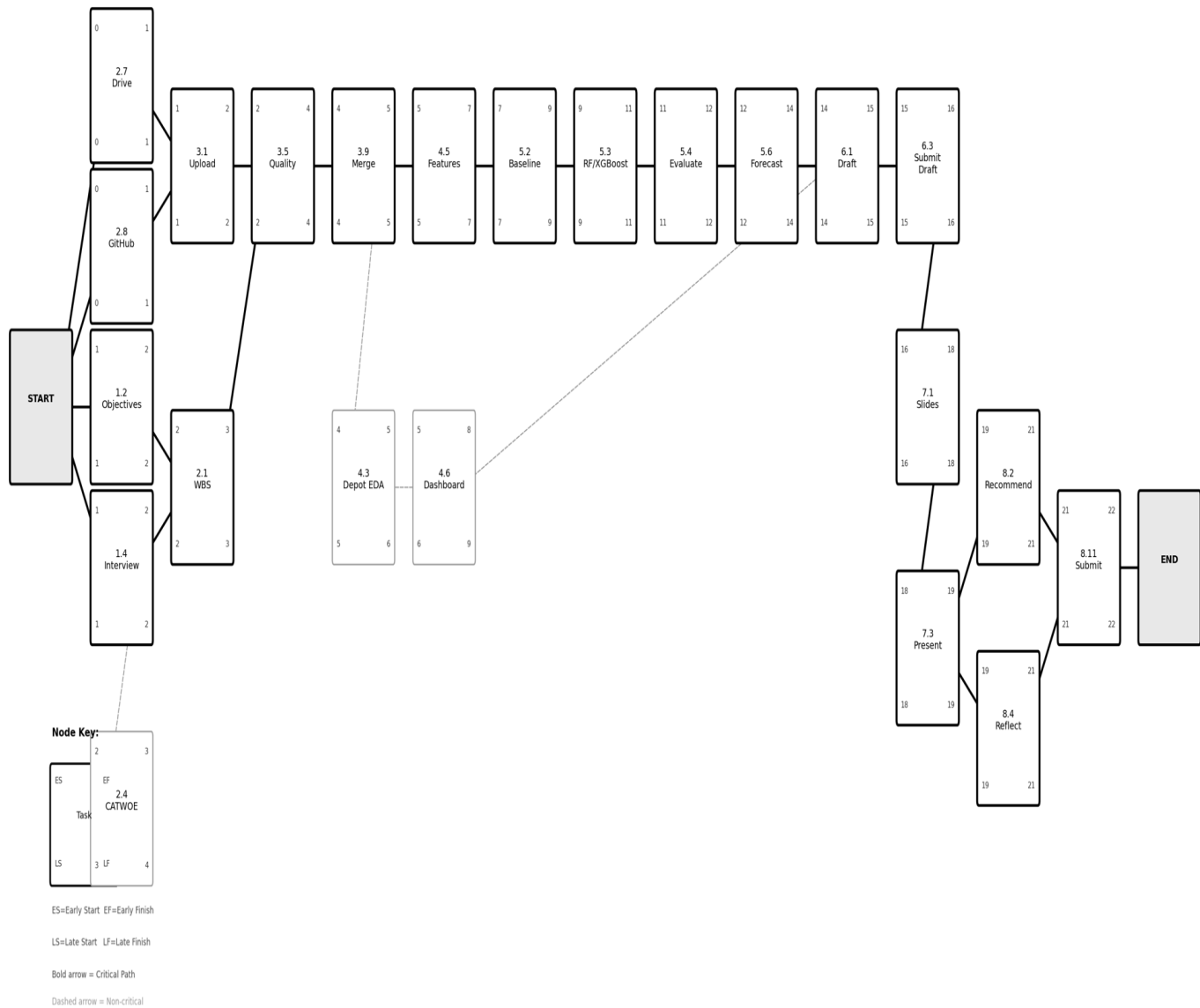
Risk Analysis

(W1987563) DR

Categorie	Risk	Description	Mitigation
Data	Missing or incomplete data	Some datasets may contain missing records or inconsistent reporting periods.	Perform data cleaning and validation before analysis
Data	Inconsistent data formats	Datasets come from different sources and formats (CSV, ODS, XLSX).	Standardise formats and perform preprocessing before analysis
Data	Limited coverage for LNER specifically	Some datasets cover all operators rather than LNER alone.	Filter datasets to isolate LNER related records
Analytics	Misinterpretation of delay causes	Delay categories may overlap or require domain knowledge to interpret correctly.	Consult rail industry documentation and definitions for delay codes.
Analytics	Incorrect assumptions in analysis	Analytical models may incorrectly infer relationships between variables	Validate findings using exploratory analysis and cross check with multiple datasets.
Analytics	Bias in dataset selection	Selected datasets may not represent the full operational picture	Use multiple complementary datasets to improve reliability
Operational	Time constraints	Limited project time may restrict depth of analysis.	Prioritise key datasets and focus on the most relevant indicators.
Operational	Technical challenges	Issues with software compatibility or dataset processing.	Use reliable tools such as Python or data analysis software with appropriate libraries.
Operational	Data integration complexity	Combining datasets from multiple sources may be difficult.	Use consistent keys such as operators, time periods, or stations for merging datasets.
External	Changes in rail industry reporting standards	Performance metrics or reporting methods may change over time.	Clearly document assumptions and data limitations.
External	External distributions	Events such as strikes or extreme weather may distort historical data patterns.	Identify and annotate unusual periods in the analysis.
External	Data availability restrictions	Some rail datasets may be removed or updated.	Store copies of datasets used in the project.

Active network diagram (MB w2066721)

Activity Network Diagram — LNER Fleet Availability Risk Project (Group 1)



Dataset Overview (RR W2073057)

Three datasets were obtained for this project to analyse the performance of the fleet. The following datasets were used for this analysis: Fleet Performance data, Fleet Delay data, and Fleet Planning data. Each of these data sets provided insight into the railway. Together, they helped to provide insight into the overall performance of the rail fleet.

Dataset 1 – 20260213 Fleet Performance (RR w2073057)

The Fleet Performance dataset contains 3,407 rows and 45 columns. Each row represents an operational incident that affected train services. These incidents include faults, engineering issues, and other disruptions that may impact fleet availability.

```
1 fleet_performance.shape  
(3407, 45)
```

Several of the variables within the dataset are useful for performing analysis of the data. For instance, the Incident Date and Incident Location variables help to reveal when and where the incidents occurred. The Total Mins and Primary Mins Sum variables indicate the number of minutes that the incidents took up. Furthermore, the Sum Canx and Train Type variables indicate the number of cancellations of that train type. Finally, variables like Cause Code, Root Cause, System, Subsystem, and Failure help to indicate the reason for the incidents to occur altogether.

1 fleet_performance.head()																						
Row	Status	Period	Incident Number	Incident Date	Headcode	Train	Incident Location	TARA Primary Delay Minutes	TARA Canx	Primary Mins Sum	***	RPE Code	Cause Code	System	Subsystem	Failure	Engineering Comments	ETC (If Required) In Progress?	PHD?	PHD Number	PTP/PHD Recommended?	
0	New	2012	192749	2026-02-12	1C15	09:52 ABERDEEN to KING'S X	LEED MILL	17.0	0.0	17.0	...	701D	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
1	New	2012	183000	2026-02-12	5308	08:34 PERMETHURK to KING'S X	F HAMPHRS	5.0	0.0	5.0	...	701A	100	Operational	NR	Regulation	Unit late off depot	NaN	NaN	NaN	NaN	
2	New	2012	191770	2026-02-12	5402	NaN	XNL	NaN	NaN	18.0	...	701D	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
3	New	2012	191716	2026-02-12	3163	04:35 DONGAHEAD to YORK (running as 04:57 (KING'S X))	DOON	10.0	0.0	10.0	...	701D	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	
4	Fixed Cause, Single Dispute Status		2012	190994	2026-02-11	1026	10:53 KING'S X to SKIPTON	DOON	11.0	0.0	11.0	...	701D	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	

For data quality, variables such as Incident Date and Total Mins have no missing values and are, therefore, reliable variables for analysis. However, some of the technical variables have moderate missing data (between 10-20% missing values). Additionally, there are several variables that have very high percentages of missing data (above 80-90%) which will likely be removed during the data cleaning process.

1: fleet_performance, lnae(), sum()/lme(fleet_performance)*500	
0	
Row Status	0.000000
Period	0.000000
Incident Number	0.000000
Incident Date	0.000000
Headcode	51.850350
Train	5.430481
Incident Location	1.985201
TADA Primary Delay Minutes	10.965396
TADA Cost	10.084746
Primary Mine Sum	0.807945
Total Mine	0.000000
Sum Cause	0.507027
Equipment Diggle	0.857189
Vehicle No.	63.780452
Unit No.	0.645729
Coupled Unit No.	88.801303
S-3 Car	2.230701
Train Type	0.675001
Train Accepted	20.780748
LNED Investigation Notes	0.907945
Hitachi Investigation Notes / SAP Info / Head Cause	0.801400
LNED Status Notes	30.861168
Diggle Mgr	9.840211
Train Mgr	0.507027
Inc Cause	0.507027
Root Cause	71.426475
Diggle Dispute Status	0.507027
Diggle Status	0.507027
LNED Fleet Status	0.000000
Report Requested From/Delay Attributed To:	
Report Requested From/Delay Attributed To:	83.924274
Master Incident Number	0.000000
Date Report Requested/Referred or Delay Closed Out	06.791899
Hitachi Station	17.816261
Tech	10.390373
TIN	0.000000
KPI Code	0.618378
Cause Code	14.587614
System	14.610005
Subsystem	14.610005
Failure	15.627584
Engineering Comments	31.853713
EFC (if Required) In Progress?	99.911946
PHRD?	100.000000
PHRD Number	99.570549
PHRD Recommended?	100.000000
dtype: float64	

Overall, this dataset is useful for analysing operational disruptions, engineering failures, and the impact that different incidents have on delay minutes and cancellations.

Dataset 2 – Fleet Delay Data from P2301 to P2611 (RR w2073057)

The Fleet Delay dataset contains 3,925 rows and 55 columns. Each row represents a delay incident affecting train services. Compared to the Fleet Performance dataset, this dataset focuses more directly on delay severity and operational impacts.

```
1 fleet_delay.shape  
(3925, 55)
```

Some of the most important variables to note in the dataset are Date, which records the date upon which each of these incidents occurred, and the variables Delay Mins and Total Mins, which indicate the severity of the delay that occurred on the given date. The variable Trains Affect indicates the number of trains that were affected by the incident, and the variable Total Canx indicates the number of cancellations that resulted of the incidents with those trains. Additionally, the variables Sig. Lates and CaSL indicate the number of late trains (or significant lateness events) and a general overall performance metric for the trains regarding the number of cancellations and lateness events, respectively. Finally, the variables Cause and Root Cause help to indicate the reason for each of the incidents, as do the variable Incident Location.

```
1 fleet_delay.head()
```

	Inc.#	Date	Status	Mile Mgr	Mile Mgr Title	Trust Mgr	Mile Mgr	Cause	Root Cause	Root Cause Descrip.	...	Region	Sect2	Day #	NR RPT	TDC1	TDC2	Region	Func1	Unnamed: 53	1246.5
0	480134270401	2022-04-01	A	MHB2	HITACHI - CLASS 801	MHB2	OGIE	MN	NaN	No Root Cause entered	...	Azuma	Hitachi - Class 801	2022-04-08	Fleet	HB	LNFR	Azuma	Azuma	NaN	NaN
1	480541270403	2022-04-03	A	MHBK	YARD - CRAIGENTRINY	MHBK	OGILS	MJ	NaN	No Root Cause entered	...	Depot	Craigentrey	2022-04-10	Depot	HB	LNFR	Depot	Azuma	NaN	NaN
2	480877720403	2022-04-03	A	MHB1	HITACHI - CLASS 800	MHB1	OGIS	MY	MYB5	Class 800/801 Couplers	...	Azuma	Hitachi - Class 800	2022-04-10	Fleet	HB	LNFR	Azuma	Azuma	NaN	NaN
3	487346270404	2022-04-04	A	MHB1	HITACHI - CLASS 800	MHB1	OGIS	M7	M703	PASSENGER DOOR	...	Azuma	Hitachi - Class 800	2022-04-11	Fleet	HB	LNFR	Azuma	Azuma	NaN	NaN
4	480234270404	2022-04-04	A	MHBA	FLEET - CLASS 911 LOCOMOTIVE	MHBA	OGIL	MB	NaN	No Root Cause entered	...	IC225	G 911 Loco	2022-04-11	Fleet	HB	LNFR	IC225	IC225	NaN	NaN

5 rows x 55 columns

In terms of completeness, this dataset is relatively clean. Most columns contain no missing values, and only the Root Cause variable has a small amount of missing data (approximately 3%). Two columns contain 100% missing values and can be removed during the cleaning process.

```
1 fleet_delay.isnull().sum()/len(fleet_delay)*100
```

Inc.#	0.000000
Date	0.000000
Status	0.000000
Mile Mgr	0.000000
Mile Mgr Title	0.000000
Trust Mgr	0.000000
Mile Mgr	0.000000
Cause	0.000000
Root Cause	3.260000
Root Cause Descrip.	0.000000
Region	0.170000
Sect2	0.000000
Day #	0.000000
NR RPT	0.000000
TDC1	0.000000
TDC2	0.000000
Region	0.000000
Func1	0.000000
Unnamed: 53	0.000000
1246.5	0.000000

CaSL	0.000000
PPM Fail	0.000000
Inc.#	0.000000
Date	0.000000
Status	0.000000
Mile Mgr	0.000000
Mile Mgr Title	0.000000
Trust Mgr	0.000000
Mile Mgr	0.000000
Cause	0.000000
Root Cause	0.000000
Root Cause Descrip.	0.000000
Region	0.000000
Sect2	0.000000
Day #	0.000000
NR RPT	0.000000
TDC1	0.000000
TDC2	0.000000
Region	0.000000
Func1	0.000000
Unnamed: 53	0.000000
1246.5	0.000000

df.columns

Because of its high data completeness and operational variables, this dataset is particularly useful for analysing delay severity, identifying common disruption causes, and understanding the operational impact of incidents.

Dataset 3 – Fleet Planning Log (RR w2073057)

The Fleet Planning dataset is structured differently from the other two datasets. It contains 52 rows and 55 columns and appears to be a planning spreadsheet rather than structured incident data. The dataset includes reporting periods, weekly planning dates, fleet availability metrics, and the number of train units required for operations.

```
1 Fleet_planning.shape
(52, 55)
```

Some of the relevant information within this dataset includes reporting periods labelled from Period 1 to Period 13, weekly planning dates, fleet classes such as 800/1, and the number of train units required to operate services. These variables provide useful context about planned fleet availability and operational capacity.

```
1 Fleet_planning.head()
```

	From Instance Score Card 2022 / 2023	Unnamed: 1	Period 1	Unnamed: 3	Unnamed: 4	Unnamed: 5	Period 2	Unnamed: 7	Unnamed: 8	Unnamed: 9	...	Unnamed: 45	Period 12	Unnamed: 47	Unnamed: 48	Unnamed: 49	Period 13	Unnamed: 51	Unnamed: 52	Unnamed: 53	Unnamed: 54
0	NaN	NaN	Week 1	Week 2	Week 3	Week 4	Week 1	Week 2	Week 3	Week 4	...	Week 4	Week 1	Week 2	Week 3	Week 4	Week 1	Week 2	Week 3	Week 4	NaN
1	NaN	NaN	2022-04-01 00:00:00	2022-04-10 00:00:00	2022-04-17 00:00:00	2022-04-24 00:00:00	2022-05-01 00:00:00	2022-05-08 00:00:00	2022-05-15 00:00:00	2022-05-22 00:00:00	...	2023-01-29 00:00:00	2023-02-05 00:00:00	2023-02-12 00:00:00	2023-02-19 00:00:00	2023-02-26 00:00:00	2023-03-05 00:00:00	2023-03-12 00:00:00	2023-03-19 00:00:00	2023-03-26 00:00:00	2023-04-01 00:00:00
2	Fleet Availability	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	800/1	Number of Units Required	90	70	70	70	70	70	70	70	...	70	70	70	70	70	70	70	70	70	60

5 rows x 55 columns

However, the structure of this dataset is more complex than the others. It uses a multi-row header format where week numbers and dates appear in separate rows, and many columns are labelled as Unnamed. This indicates that the dataset was likely designed for manual reporting rather than direct analysis. As a result, significant restructuring and cleaning will be required before it can be used effectively in analysis.

Despite these challenges, the dataset may still provide useful context regarding fleet planning and operational demand.

How the Datasets Work Together

These three datasets provide different perspectives on the operations of the railway. The Fleet Performance and the Fleet Delay datasets contain information regarding the incidents that occurred on the railway system. Additionally, the Fleet Planning dataset provides information regarding the planning of the railway system and its capacity.

By analysing these three datasets together, it is possible to investigate the relationship between the engineering failures of the railway and the planning of the system. This information can help to determine if there are any trends or patterns regarding delays, cancellations, or availability of the railway system's fleet. Further exploratory data analysis will be carried out to determine how these datasets can best be used to support modelling and predictive analysis.

Summary (MC W2068659)

Fleet Delay Dataset (MC W2068659)

The Fleet Delay dataset contains information regarding delay incidents between periods P2301 and P2611. The dataset contains fields for information such as the date of the incident, the cause of the delay, the root cause of the delay, and the number of minutes that the delay took. The dataset is complete overall with few missing values. Some columns contain many missing values and are believed to be artefacts of the original spreadsheet. There were no duplicate data records within the dataset. This data contains the main information regarding fleet delays.

Fleet Performance Dataset (MC W2068659)

The Fleet Performance dataset contains information regarding fleet incidents and the minutes during which the vehicles were delayed from performing their intended tasks. Fields regarding variables like delay minutes, incident dates, and other general notes regarding each of these incidents are available within the dataset. Some fields contain missing values, such as those related to the engineering and investigation of these incidents. However, the general information regarding the incidents itself is complete and available for analysis regarding the performance and reliability of the fleet overall.

Fleet Planning Log Dataset (MC W2068659)

The Fleet Planning Log dataset contains information regarding the planned deployment of the company's vehicles. The dataset has several columns with planning data but has many missing values. Some of the columns have no named header, and there are also some duplicate rows in the dataset. However, this dataset does reveal some information regarding the company's planned fleet deployment.

Common Join Key Between Datasets (MC W2068659)

Upon initial inspection, the two datasets do not appear to have any columns that are similarly named. However, both datasets include columns related to the time in which the data was reported, which can be utilised to link the two tables together. This will allow for the tables to be joined in a manner that creates the master database for analysis.

Cleaning Datasets (MC W2068659)

Cleaning Delay dataset

Stage	Rows	Columns
Before Cleaning	3925	55
After Cleaning	3925	53

Removed column 1246.5

Almost entirely missing values in this column indicate its creation as an artefact of the spreadsheet created during the Excel export process. Thus, because this column does not contain any analytical information, it was removed from the dataset.

Converted Date column to datetime format

This step ensured that the date column was properly formatted for analysis, as well as for the merging of other data with this data.

Handled missing values in Equip.

There was one missing value for the equipment column. Replacing the missing value with “Unknown” ensured that the record could be included in the database without introducing a null value.

Removed duplicate rows

Any rows that contained identical information to another row in the database was removed.

Cleaning Performance dataset: (MC W2068659)

Stage	Rows	Columns
Before Cleaning	3407	45
After Cleaning	3407	45

Converted Incident Date column to datetime format

The column Incident Date was converted to a proper datetime format to allow for the merging of this dataset with others.

Renamed Incident Date column to Date

The column Incident Date was renamed to Date to ensure consistency with the delay dataset. This will allow for easy merging of these two different datasets into one master data frame.

Handled missing values in key operational columns

Several of the operational columns (such as Headcode, Train, Incident Location and Root Cause) contained missing values. These values were replaced with “Unknown” to remove any potential issues caused by these missing values.

Handled missing values in investigation and engineering fields

The investigation and engineering related fields (such as Engineering Comments, LNER Investigation Notes and Hitachi Investigation Notes / SAP info / Root Cause) also contained missing values. These fields were replaced with the value of “Not recorded” to indicate that no investigation notes were recorded for these incidents.

Removed duplicate rows

Any rows that contained duplicate information was removed to ensure that each incident was represented only once in the database.

Cleaning Planning dataset (MC W2068659)

Stage	Rows	Columns
Before Cleaning	52	55
After Cleaning	43	14

Removed spreadsheet artefact columns

The columns that were labelled “Unnamed” were artefacts of the formatting software (Microsoft Excel) that was used to create the spreadsheet.

Handled missing values using forward fill

Cells with missing values were filled in using forward-fill methods to ensure that the planning data remained intact.

Removed duplicate rows

Rows that contained identical data to another row were removed to ensure that each row represented a unique planning task.

Reasons for Not Merging the Datasets (MC W2068659)

When attempting to merge the two datasets, inconsistencies and missing data for several of the variables were introduced.

Some of the records could not be accurately matched between the two datasets, which could lead to incorrect and duplicated data being introduced into the analysis.

Furthermore, the merge also led to a reduction in the number of available records for analysis due to the missing IDs for several of the records in the datasets.

To ensure the integrity of the data and to avoid any noise or bias being introduced into the analysis of the data, the datasets were analysed separately.

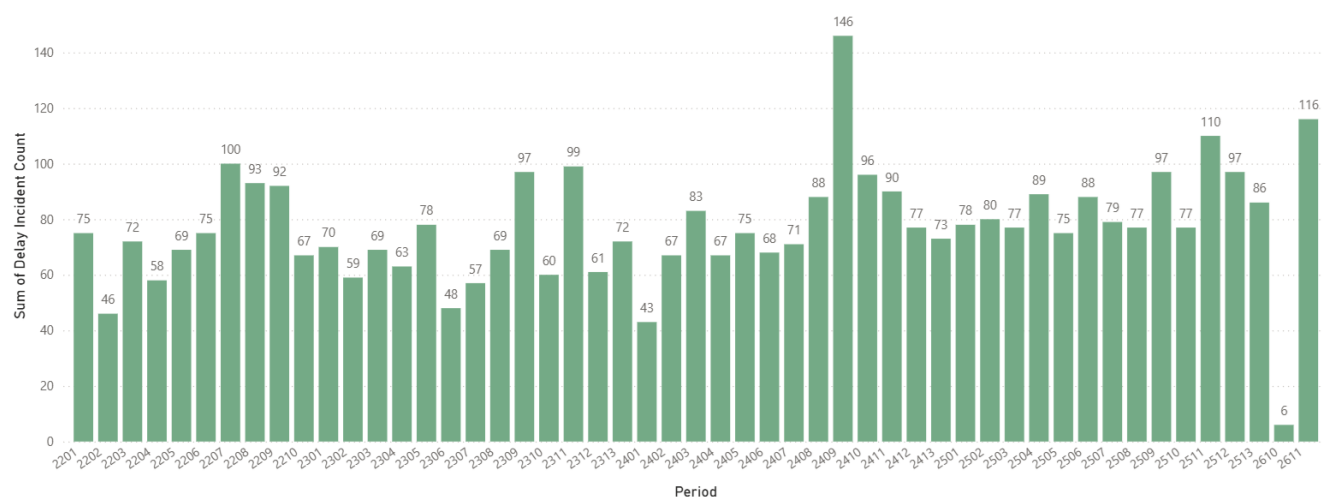
By analysing the datasets separately, it was possible to utilise all the data while ensuring the reliability of the variables within the datasets.

Exploratory Data Analysis

Figure 1: Fleet Delay Incidents by Period (P2201 – P2611)
SJ(W2117206)

Stage	Rows	Columns
Before Cleaning	52	55
After Cleaning	43	14

Fleet Delay Incidents By Period (P2201 -P2611)

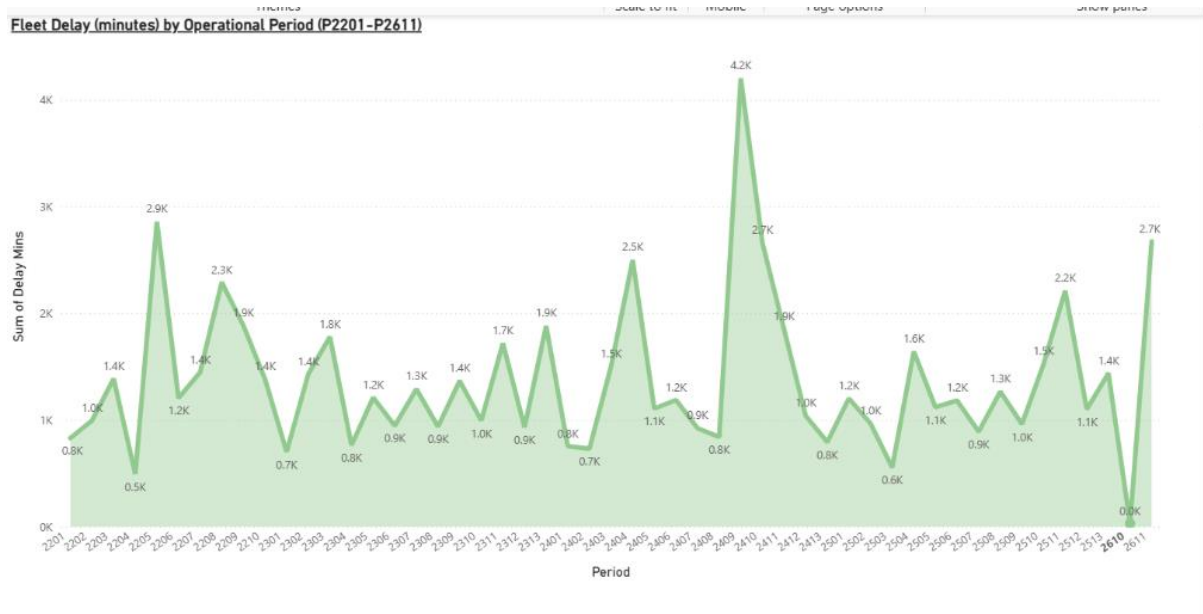


The clustered column chart above shows the number of fleet delay incidents recorded across the operational periods P2201 to P2611. Each column represents the total number of delays within a specific operational period. Therefore, comparisons can be made between operational reliability over time.

The chart implies that most periods experience between 60 to 100 incidents each period. However, there are noticeable spikes. For instance, in period 2409, there was a steep increased compared to period 2408, where P2408 had 88 incidents in comparison to 146 in Period 2409. This indicates that during this period, there was increased disruption which resulted in fleet delays.

Delay incidents occur consistently which can be seen in the graph. The fact that these delay incidents occur so often within an operational period suggests that disruptions are regular within LNER's network. This is why it is essential for LNER to identify periods with higher delay frequencies. Since there are periods with frequent delays, this shows that delays are not isolated so repeated disruptions may potentially impact fleet availability and increase the risk of timetable instability.

Figure 2: Fleet Delay (minutes) by Operational Period (P2201 – P2611)
SJ(W2117206)



The line chart above shows the trend in fleet delay minutes across operational periods P2201 to P2611. Each point in the chart represents the sum of delay minutes which were recorded within a specific operational period. Having this line chart, comparisons between each data point from a particular operational period can be observed.

While most periods show a sum delay time of around 800 to 1500 minutes, noticeable spikes can be seen. For example, the total delay time across period 2410 exceeded 4000 minutes compared to period 2409 where the total delay time was only around 800 minutes.

Spikes in the chart signifies periods of high operational disruption. These spikes could stem due to reliability issue, operational constraints, or faults within LNER's infrastructure. These spikes in delay minutes during a certain operational period implies that some high operational disruptions are more severe than others. This means that increased pressure on fleet scheduling and fleet recovery may be created. It is very important for LNER to analyse the trends between delay times across time because the increases could signal more pressure on fleet availability and higher risks of service disruption.

Figure 3: Operational Issues by Late Started Events (Depot)
SJ(W2117206)

Abbreviated	Actual
IEP	Doncaster Carr
BN	London Bounds Green
NL	Leeds Neville Hill
EC	Edinburgh Craigentenny
HT	Newcastle Heaton
INV	Inverness
AC	Aberdeen Clayhills
FMP	London Ferme Park

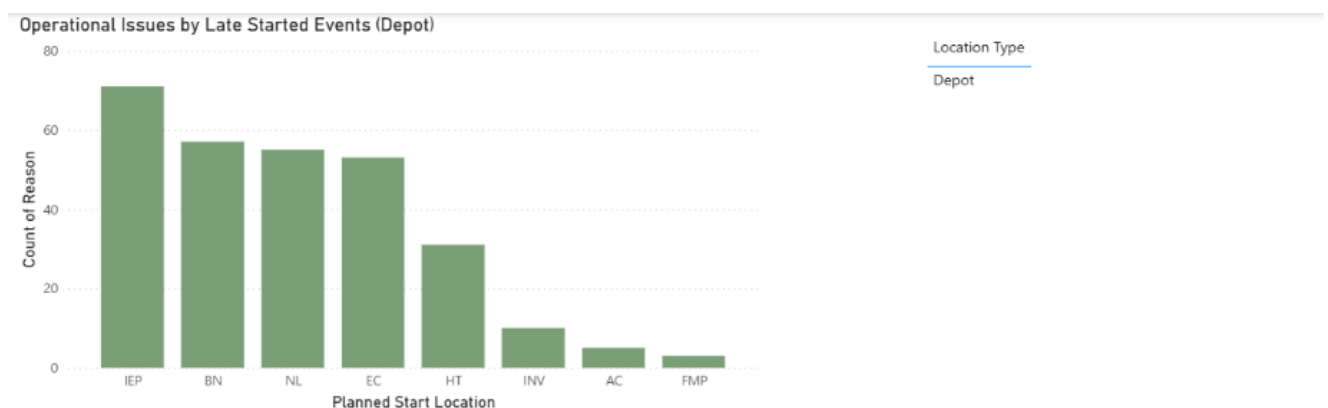


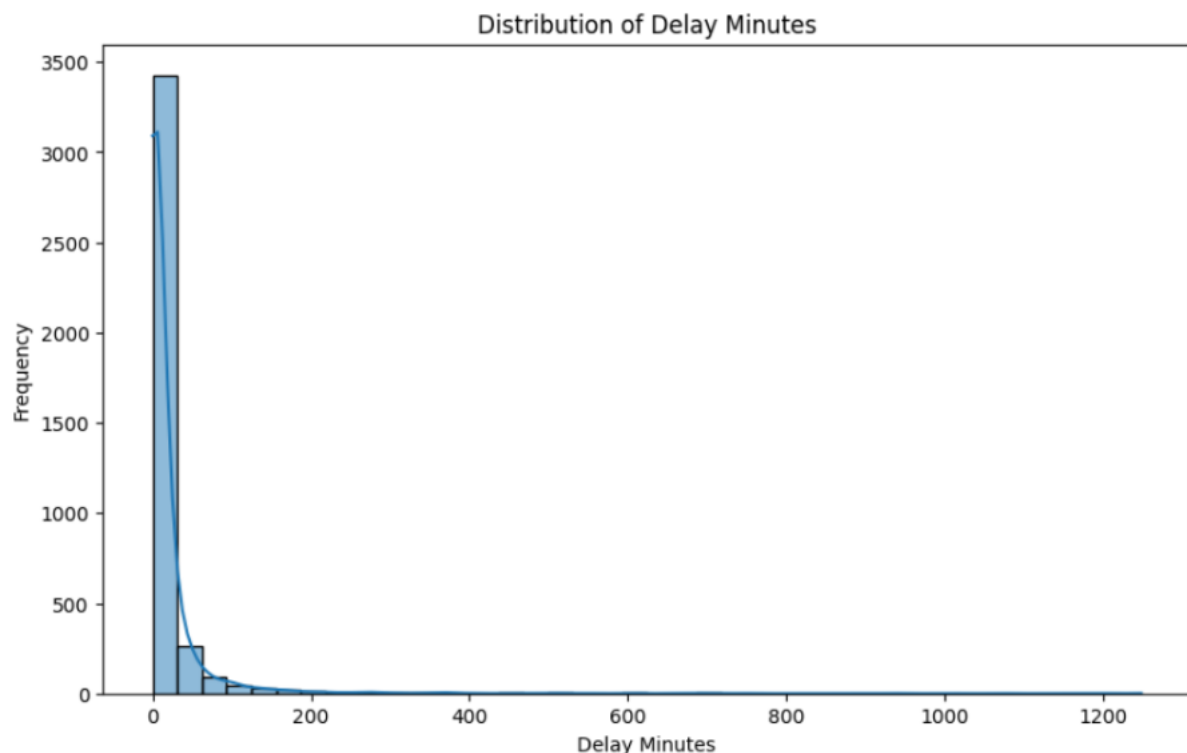
Figure 3 shows a clustered column chart which shows a depot-level breakdown of operational issues. These issues result in late fleet starts from the start location. Each column in the chart represents the total number of issues between each depot.

The chart has been filtered so that only the late started issues occurring at depot-level are being tracked. The locations have also been abbreviated.

Looking at the chart, we can see that IEP, BN, NL, and EC have the highest incident counts. Since these depots have the highest incidents counts, this suggests that these depots should be prioritised in fleet availability monitoring or early warning systems. This is because they may experience higher operational pressure. Whereas, HT, INV, AC, and FMP show lower incident counts. This implies that the pressure for fleets to be rolled out may be limited to certain key maintenance location.

LNER can track which depots have the most frequent incidents counts. This can help in solving how to improve fleet availability and reducing service disruptions since operational improvements and maintenance interventions can be prioritised depending on location.

Figure 4: Distribution of Delay Minutes (RR W2073057)

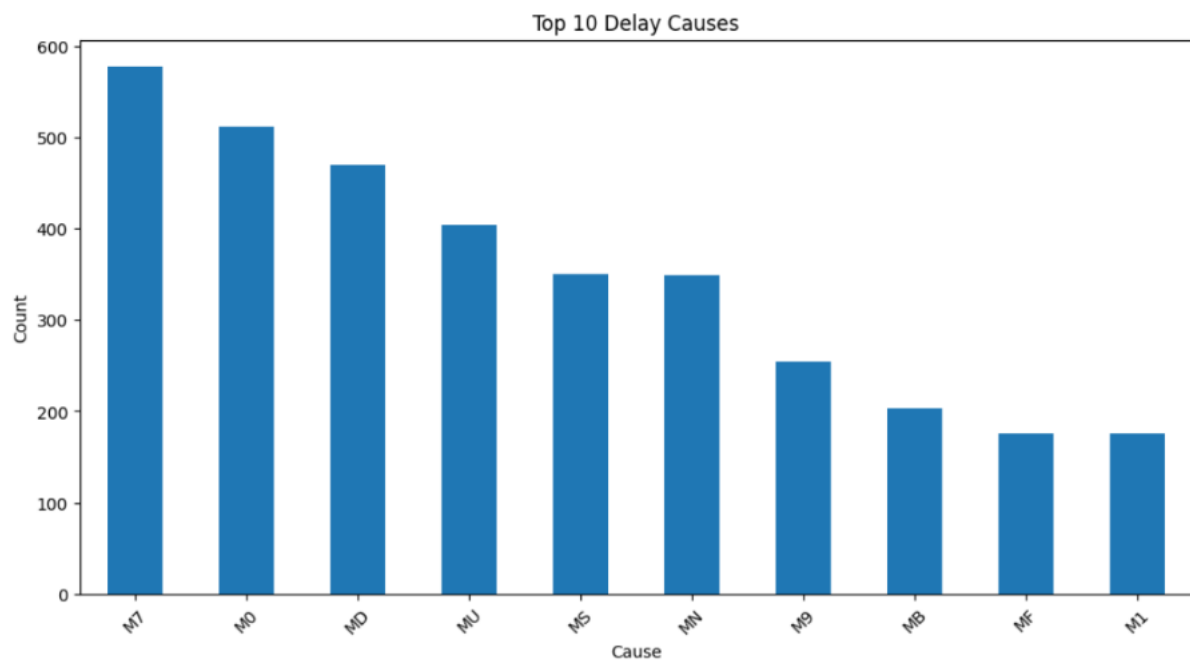


The distribution of the delay minutes was analysed to determine the severity of the delays across the LNER network. The data from the histogram demonstrates that the bulk of the delays with low minutes of delay are highly right skewed by a few incidents with long delays.

Most of the delays experienced by LNER are of a low level of severity and can easily be managed by the company. However, there are occasionally severe disruptions to the services provided by LNER. These longer delays are usually the result of complex issues that impact the services, such as infrastructure problems, weather or failures of the trains while in operation.

Furthermore, as there are very few spare trains for LNER to accommodate for disruption to services, any delay with high minutes of delay has a significant impact on the company and its network of closely connected rail services. Thus, preventing these long delays is of more importance to LNER than reducing the number of delays with low minutes of delay. Therefore, methods that aim to both provide insight into the causes of these long delays will be of more importance to the company than methods that aim to manage the less impactful delays.

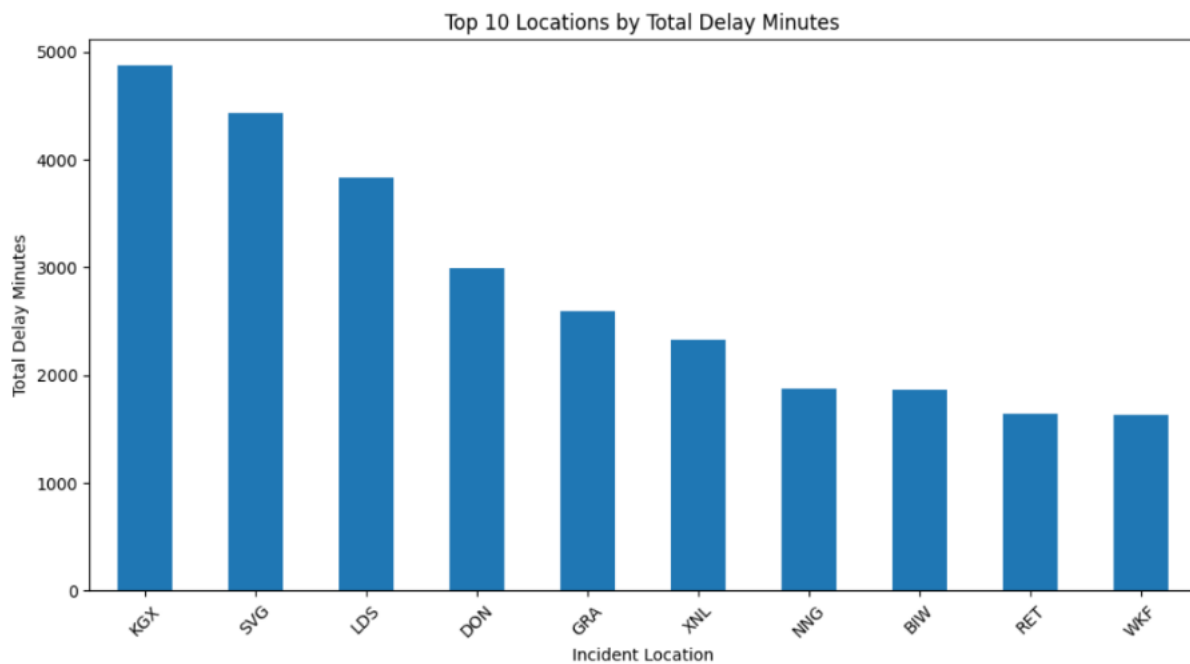
Figure 5: Top 10 Delay Causes (RR w2073057)



Cause Code	Example Desc	interpretation
M7	Passenger Door	Door fault/ passenger interference
M0	Driver/ Traincrew	Staff – related operational delays
M8	Toilet and water systems	Onboard equipment issues
MD	Diesel Engine / Mechanical	Mechanical failures
MU	No root cause entered	Unrecorded data
MN	Brake control unit	Technical braking issues
MF	Doors blocked	External obstruction
MS	Unit available	Fleet availability issues
M9	Safety of the line	External safety reports
MB	No root cause entered	Unrecorded data

Analysis of the causes of these delays indicates that both mechanical issues (such as issues with train doors, braking systems, or engines) and operational issues (such as issues with the crew or the availability of these train units) are present in the LNER. However, many of the incidents related to these delays have no recorded cause for that delay altogether.

Figure 6: Top 10 Locations by Total Delay Minutes (RR w2073057)



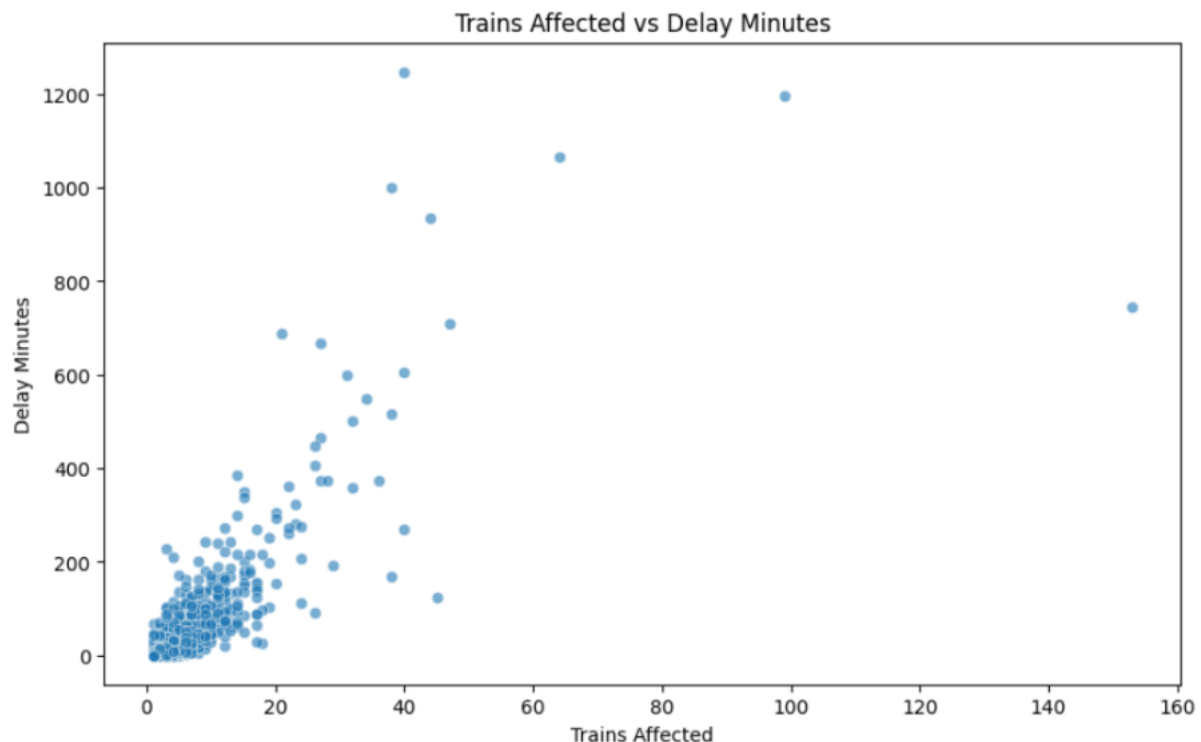
Abbreviated Location Name	Actual Location Name
KGX	London King's Cross
SVG	Stevenage
LDS	Leeds
DON	Doncaster
GRA	Grantham
XNL	Neville Hill (Leeds)
NNG	Northallerton
BIW	Berwick-Upon-Tweed
RET	Retford
WKF	Wakefield

The abbreviation location name table represents the names of the locations of the incidents as they are displayed on Figure 6, which correlates to the actual location name of these incidents on the map. This table was created to increase the interpretability of the map to demonstrate the patterns of the operational issues of the LNER geographically. This table would be important for the LNER to recognize which areas of the LNER have these operational issues.

The total delay minutes for each incident location were calculated to determine the impact of the incidents on the overall network. The chart demonstrates that a small number of locations have contributed to many total delay minutes, indicating that the incidents in those locations have had a significant impact upon the LNER network overall.

It should be noted that the location codes shown (e.g. abbreviations) are internal operational identifiers used within the dataset. As a result, the analysis focuses on identifying relative disruption levels between locations rather than interpreting specific station names, while still highlighting the areas of highest operational risk.

Figure 7: Trains Affected Vs Delay Minutes (RR w2073057)

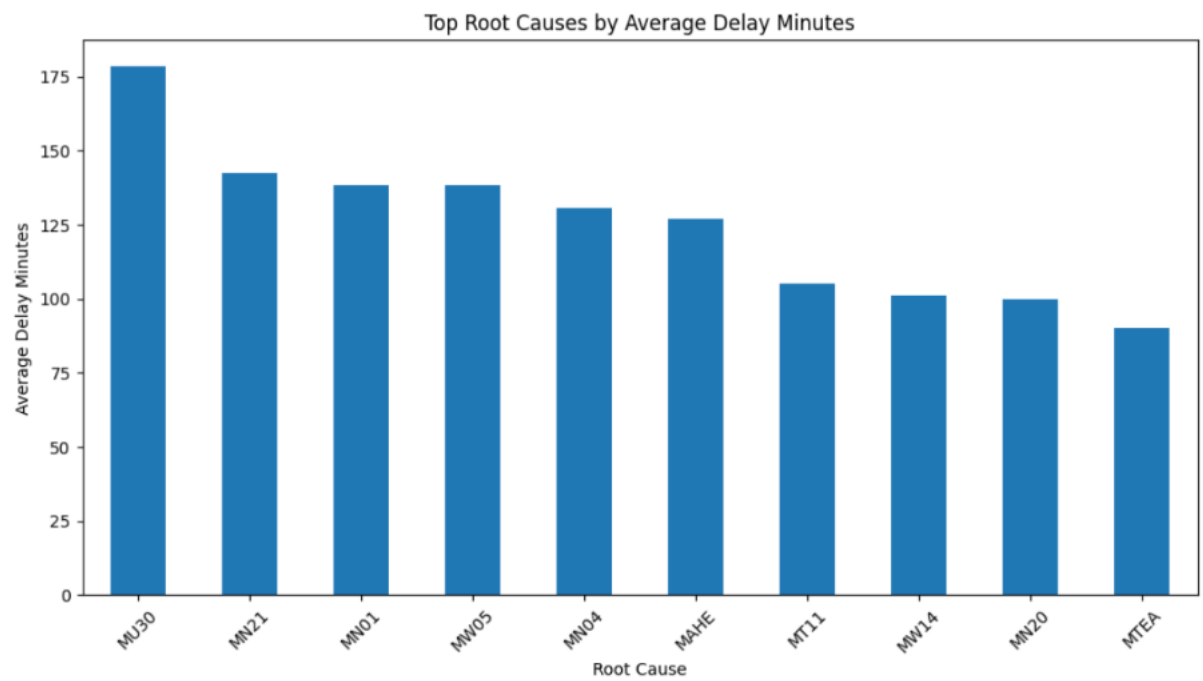


A scatter plot was used to investigate the relationship between the number of trains affected by incidents and the total delay minutes for the LNER network. The scatter plot illustrates a positive relationship between these two variables - the more trains that are affected by an incident, the more delay minutes will be incurred on the LNER network.

The reason for this relationship is due to the way in which the LNER operates its trains and its control centre. The more trains that are affected by an incident, the more difficult it is for the control centre to efficiently recover the LNER timetable. This is due to the limited number of spare trains that are available for the LNER to deploy to cover for the delayed trains.

These findings highlight that incidents affecting many trains are more operationally critical, as they contribute more heavily to overall network pressure. This supports the modelling approach used in the project, where variables such as trains affected play an important role in identifying high operational pressure situations.

Figure 8: Root Cause Severity (RR w2073057)

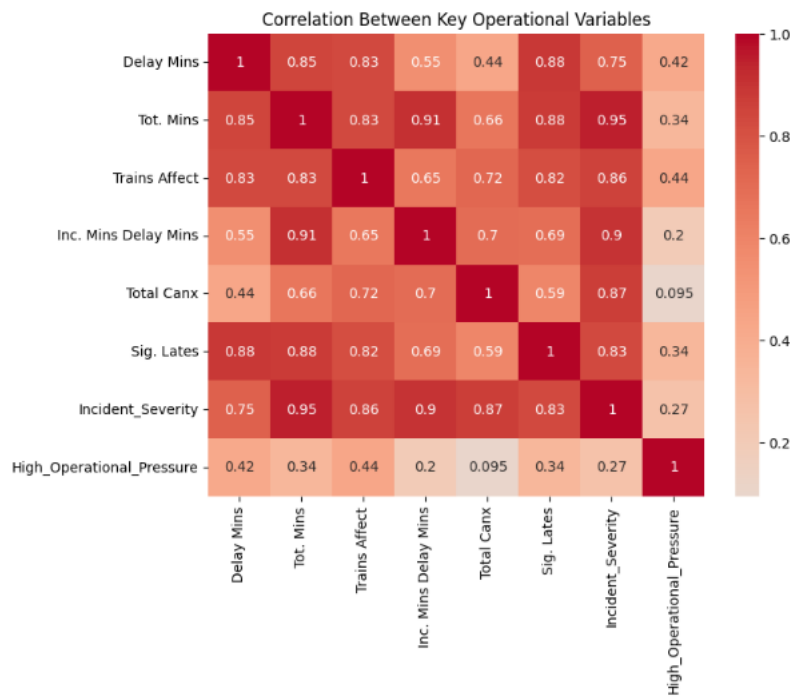


Code	Meaning
MU30	Depot OHLE
MN21	WSP Electronics/ Cables
MN01	Brake Blocks/ Pads
MW05	Ice/ Snow buildup or pantograph damage
MN04	Callipers/ actuators / cylinders
MT11	On-board HABD including cables
MAHE	Electrical solenoid valve
MW14	Train striking ice/snow damage
MN20	WSP probe
MTEA	VMS control unit

The root causes identified within the dataset are represented by internal fault codes (e.g., MU30, MN21). Each of these codes represents a specific type of technical issue with the fleet members. For instance, MU30 indicates that there were issues with the depot overhead lines, MN01 indicates that there were brake block/pad issues, and MW05 indicates that there were issues caused by ice and snow damaging the pantograph. Each of these codes can be utilized to determine which root cause creates the longest delays for the fleet as a whole. To understand which of the root causes creates the most delay for the fleet members, the average delay minutes for each root cause was calculated.

The results of these calculations allow for the understanding of which technical faults with the fleet members lead to the highest amount of delay for those members on the routes.

Correlation Analysis (RR w2073057)



A correlation heatmap was utilized to visualize the relationships between key variables, such as delay minutes, total delay minutes, trains affected, and the severity of the incidents that occurred on the LNER network.

The heatmap shows strong positive correlations between several disruption metrics. For example, delay minutes and total delay minutes are closely related, reflecting how severe incidents generate larger network-wide delays. Similarly, the relationship between trains affected and delay minutes is also strong, suggesting that incidents affecting multiple trains tend to create more delays across the network.

In the context of LNER's operations, such relationships between the different measures of delay highlight the interconnected nature of the problem. These relationships help to identify the causes of delay, which can then help to identify which features will be utilized in the model to identify periods of high risk of delay for the company's trains.

Exploratory Data Analysis Conclusion

Overall, the exploratory data analysis highlights the incident and delay patterns that occur across LNER's network which are beneficial to understand. Figures 1 and 2 shows that there are always delays occurring across operational periods; we can see that some operational periods experience more delays and delay minutes compared to others. Figure 3 reinforces the fact that there are delays by showing the incident counts that happen depot wide. In Figure 3, we see that some depots have higher numbers of late start incidents. These figures also show that these delays which happen are not evenly distributed, as some depots have higher late start counts and over certain operational periods, the number of delays and the minutes of these delays vary. This shows that the severity of these delays not only occur frequently, but delays also start from their respective depots.

Figure 4 reinforces the fact that the distribution of the delays and delay minutes are uneven. Looking at Figure 4, we can infer that most delays are not severe, however, we can see that severe delays can occur occasionally which contributes to the uneven distribution of delays and delay minutes. Figures 5 and 8 shows what causes these delays, which shows that both mechanical and operational faults contribute to delays, with certain root causes leading to more severe delays and increased delay minutes.

Figure 6 shows the geographical locations which have the highest delay minutes. This suggests that delays in these specific locations have a bigger impact on LNER's network. Figure 7 shows a positive correlation between number of trains affected and the delay minutes. This links to Figure 6 because it highlights the fact that disruptions can circulate across LNER's wider network.

In conclusion, the information gathered across the Exploratory Data Analysis demonstrates that fleet availability risks have many factors to consider. This includes frequency of delays, delay minutes, severity of delays, and the operational pressures at specific depots and locations. Understanding these issues and providing solutions to each of these factors can eventually lead to a decrease in delays and fleet availability across LNER's network.

Data Preparation, Modelling, and Evaluation (RRw2073057)

The approach to modelling was based on the decision-making process of the LNER. It was felt that the model should not just be able to explain the disruptions that had occurred in the past but also be able to identify periods of high operational pressure. A classification approach to machine learning was felt to be the most appropriate algorithm to utilise, given the nature of the data and the operational pressure rule.

During the initial model building stage, the model achieved very high accuracy scores. However, upon investigation, it was revealed that these high scores were the result of data leakage.

By removing these features that had revealed a relationship with the target variable, and reiterating the model build process, a more realistic model was created.

Several different models were created during this analysis rather than just creating one model. This was done to ensure the robustness of the results obtained from the models. Furthermore, it was also of interest to compare the interpretability of these models and the complexity of explaining each of the models. Within the field of data science, it is generally always better to create several different models for a specific task. This is especially true in the context of the real world and application of these identified models within the operations of the LNER.

Distribution of Operational Pressure (RR w2073057)

A binary target variable called High Operational Pressure was created to classify incidents into two categories: normal operational pressure and high operational pressure.

The distribution plot shows how incidents are split between these two classes. High operational pressure incidents represent the most severe disruptions in the dataset and are the focus of the predictive modelling stage.

```
High_Operational_Pressure
0      2965
1       960
Name: count, dtype: int64
Threshold used: 26.0
```

Feature Engineering (RR w2073057)

The feature engineering process transforms raw data into useful information for the machine learning models.

A threshold of 26 minutes was used to determine high operational pressure. This threshold was selected based on the data of the number of minutes of delay of all incidents. This threshold allows for the model to consider only the incidents with significant delays. Furthermore, using this threshold allows for the separation of the data into two classes with a sufficient number of observations in each class. This threshold balances the classes while ensuring the outcome reflects the operational pressure of the incidents.

All of the categorical variables are converted into a numerical format through one-hot encoding to allow the machine learning models to understand these variables.

Several variables related to the time of the incidents were created, such as the date and the day of the week of the incidents.

All binary variables, such as late and cancellation flags, are also considered important variables to include in the models.

Several variables related to the delay of the incidents were considered for inclusion in the model but were later removed due to the potential for data leakage. These variables were removed to ensure that the machine learning models only relied on variables available prior to the events of high operational pressure.

The results of the feature importance show that there are only a few variables that have significant importance in determining whether an incident will result in high operational pressure. The number of trains affected by the incident and the late flags are the two most important variables. This indicates that the model is not complex but relies upon a few variables to make its predictions.

The feature engineering stage focused upon creating variables that could be used to make forward-looking and generalizable predictions about operational pressure.

Assumptions and Modelling Decisions (RR w2073057)

Several assumptions were made in the construction of the models.

The first of these assumptions was the use of a 26-minute threshold to indicate high levels of operational pressure. This threshold was created based upon the distribution of the data itself and indicated the threshold at which delays were significant enough to be included in the binary classification model.

Only features that existed prior to the outcome of the given flights were used in the construction of the models. Any variables that indicated the outcome of the flights or were calculated because of the delay of those flights were removed from the models to ensure that the models reflected the likelihood of delays before those flights occurred.

Lastly, variables like the number of trains that were affected were included in the models. These features are highly predictive of the outcome of the given disruptions but are also indicators of the extent of that disruption itself. While not ideal, these features were included in the models to reflect the extent of disruptions in a way that was understandable to non-technical audiences.

One – Hot Encoding (RR w2073057)

Some variables in the dataset, such as root cause and incident location, are categorical rather than numeric. Machine learning models cannot work directly with text categories.

To address this, one-hot encoding was used to convert categorical variables into binary indicator variables. Each category is represented as a separate column indicating whether that category is present for a particular incident.

This allows the machine learning models to incorporate categorical operational factors into the prediction process.

```

1 #one hot encode categorical variables
2 X = pd.get_dummies(X, drop_first=True)
3
4 print(X.shape)
5 print(X.head())

```

```

*** (3925, 946)
   Delay Mins  Inc. Mins Delay Mins  Trains Affect  EQ Mins  Capes  Pines  \
0      0.0      0.0      0.0      0.0      1      8.0      0.0      0.0
1      2.0      0.0      0.0      0.0      1      2.0      0.0      0.0
2      0.0      0.0      0.0      0.0      1      6.0      0.0      0.0
3      3.0      0.0      0.0      0.0      1      3.0      0.0      0.0
4     181.0     143.0     143.0      0      0     584.0      2.0      1.0

   Calvs  FTS  Full Canx  Part Canx  ... Equip._UNKNOWN Equip._Undefined  \
0      0.0      0      0.0      0.0  ...      False      False
1      0.0      0      0.0      0.0  ...      False      False
2      0.0      0      0.0      0.0  ...      False      False
3      0.0      0      0.0      0.0  ...      False      False
4      0.0      0      3.0      0.0  ...      False      False

   Equip._Unknown  Equip._VARIOUS  KPI_701D  KPI_750  NR KPI_Fleet  \
0      False      False      True      False      True
1      False      False      False      False      False
2      False      False      True      False      True
3      False      False      True      False      True
4      False      False      True      False      True

   Region_Depot  Region_IC225  Region_Other
0      False      False      False
1      True      False      False
2      False      False      False
3      False      False      False
4      False      True      False

[5 rows x 946 columns]

```

Predictive Modelling (RR w2073057)

Three different classification models are evaluated in this report including the Decision Tree, Random Forest and Gradient Boosting models. These models were chosen mainly due to the desire to find a balance between model interpretability and accuracy. A Decision Tree model was created first. Ensemble methods, such as Random Forest and Gradient Boosting, were created after to provide further robustness and generalisation to the model.

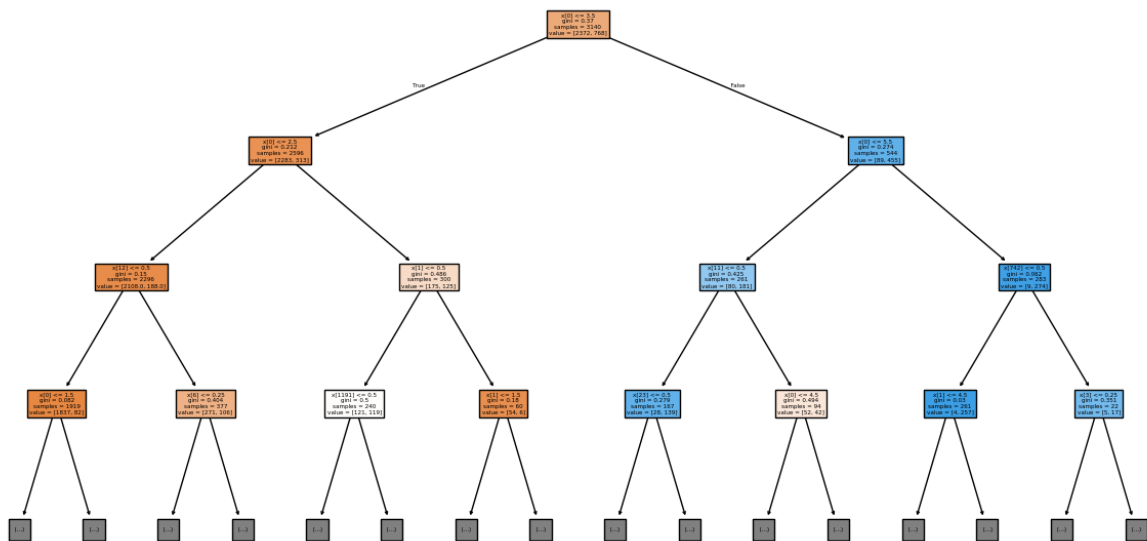
Decision Tree Model (RR w2073057)

The Decision Tree model was selected as a baseline for its simplicity and easy interpretability. It provides a clear visual representation of how features influence predictions.

The model was able to perform with an accuracy of about 88.5%. Nonetheless, this increased accuracy may be a symptom of overfitting; decision trees can be sensitive to noise, thus lacking the ability to generalise.

Furthermore, a recall of around 71% was experienced for high operational pressure events, signifying that the model performed worse in identification. This suggests that some high-risk events are missed (i.e., incorrectly classified as low risk).

The decision tree structure below shows how the model makes a decision, based on key thresholds in features such as trains affected. Although this model is more fragile than ensemble methods; this interpretability at least helps us to understand its operations.



Overall, the Decision Tree serves as a useful baseline model, providing insight into how operational features influence predictions. However, its tendency to overfit limits its reliability compared to more robust ensemble methods such as Random Forest. This limitation motivates the use of more robust ensemble models, such as Random Forest, which improve generalisation and reduce overfitting.

Random Forest Model (RR w2073057)

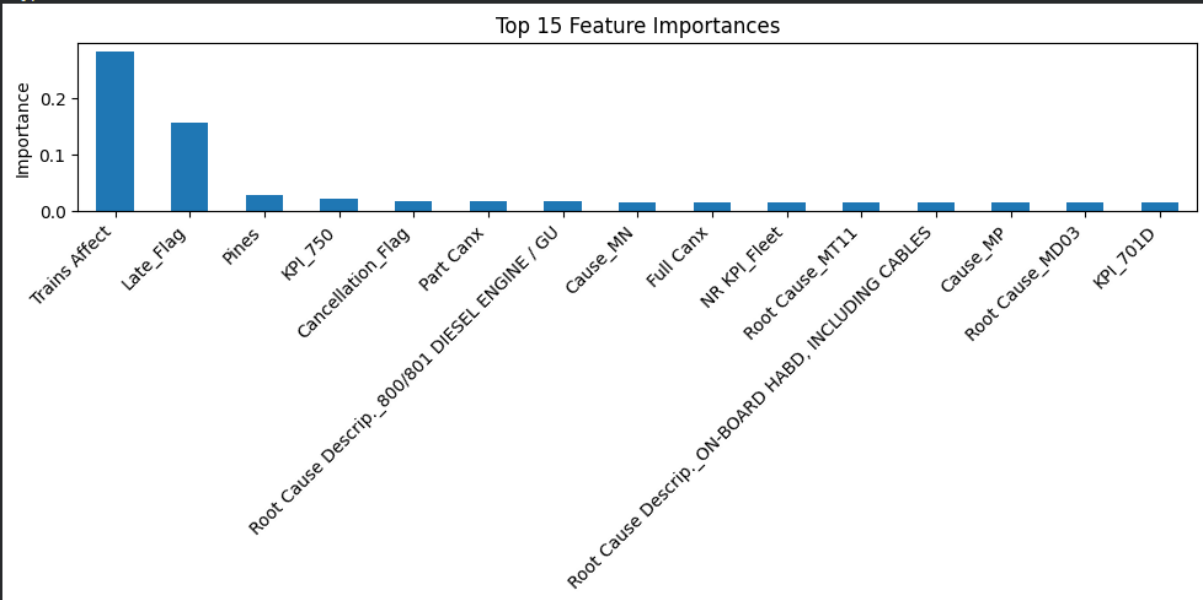
The Random Forest model was chosen as the main model for the project as it presented itself as a strong performer for the project. Random forests are a model composed of multiple decision trees created using the bagging method to reduce the variance of the models.

The accuracy of this model was approximately 85.5%, which is slightly less than that of the Decision Tree model. However, random forests tend to exhibit less overfitting and have better generalisation of the data.

The model had a recall of approximately 84% for detecting high levels of operational pressure. Thus, the model correctly detects the majority of instances of high levels of operational pressure. While the precision of the model is relatively low, this is expected for identifying such high-risk events for the railway.

In the context of operational railways, false negatives are more costly than false positives; hence, the model was specifically trained to minimise false negatives.

The importance of the features within the model indicated that features related to trains affected, late-running services, and cancelled services are the main predictors of operational pressure within the railways. These features enable the monitoring of the railway for potential failures.



The feature importance chart shows that “trains affected” and “late-running services” are the most influential predictors of operational pressure. This indicates that larger disruptions are strongly associated with high-pressure events. However, these variables are treated as proxy indicators rather than direct causes, as they reflect observable operational conditions rather than future outcomes.

Overall, the Random Forest model was selected as the final model as it provides the best balance between performance, generalisation, and interpretability. In a real-world railway context, prioritising recall is important, as failing to identify high-pressure events can lead to significant operational disruptions. Consistent performance across cross validation folds further demonstrates the model's stability.

```
Random Forest Accuracy: 0.8547770700636943

Confusion Matrix:
[[510  83]
 [ 31 161]]

Classification Report:
      precision    recall  f1-score   support

     0       0.94      0.86      0.90       593
     1       0.66      0.84      0.74       192

   accuracy      0.80      0.85      0.85      785
  macro avg      0.80      0.85      0.82      785
 weighted avg      0.87      0.85      0.86      785
```

Overfitting Check (RR w2073057)

Comparing training and testing accuracies tells us if the model is overfitting. The accuracy on the training set was approximately 86.2%, while the accuracy on the testing set was approximately 85.5%, giving a difference of approximately 0.7%. This means that the model is not too overfitted and is able to generalise well.

Moreover, we apply 5-fold cross-validation on the data to ensure optimal results. The performance of the model on all the folds was similar indicating that the results are not dependent on the train-test split.

In general, overall, this random forest model provides a good balance between performance and generalisation, thus suitable for a real-world application.

```
Train Accuracy: 0.8624203821656051
Test Accuracy: 0.8547770700636943
Gap: 0.007643312101910826
```

```
CV Scores: [0.8522293 0.86242038 0.83566879 0.85350318 0.84203822]
Mean CV Accuracy: 0.849171974522293
Std CV Accuracy: 0.009347228702086183
```

Model Comparison (RR w2073057)

Several different machine learning models were trained to predict high operational pressure, including Decision Tree, Random Forest and Gradient Boosting classifiers.

The model evaluation used recall for occurrences of high operational pressure, as underestimations of these can cause disruptions in operations. The F1 score was used instead of accuracy to account for class imbalance in the dataset.

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1 Score (Class 1)	Key Strength	Key Limitation
Decision Tree	88.5%	0.80	0.71	0.75	Interpretable, Simple	Overfitting, Weak generalisation
Random Forest	85.5%	0.66	0.84	0.74	Best balance, High recall	Slightly Lower precision
Gradient Boosting	~ 90 – 91%	0.84	0.77	0.80	Highest Accuracy	Less interpretable, Complex

The accuracy of the Gradient Boosting Model is highest (approx.) 90–91% and good overall. While it may deliver superior performance, XGBoost is also more complex to compute and

interpret than Random Forest. Therefore, it isn't preferable for operational deployment, especially where model stability and transparency are important.

Overall, The Random Forest model was ultimately selected as the final model as it provides the best balance between performance, generalisation and interpretability. In real-life, railway context, recall is of great importance otherwise operational disruptions due to failure in identifying high-pressure events can occur.

Modelling Justification and Alternatives (RR w2073057)

Gradient Boosting provides the highest accuracy, but it is computationally very expensive and less interpretable. Despite their simplicity, Decision Trees were overfitting and unreliable on unseen data.

The final model selected was Random Forest, identified for its performance while preventing overfitting of the model. The model also had a higher recall for high operational pressure events, making it more effective in an operational context where identifying high pressure situations is critical.

Limitations of the Model (RR w2073057)

Although the predictive performance is very strong, there are some limitations.

In the first place it uses a small number of dominant features as indicators (e.g. number of affected trains, lateness indicators). Although these variables are highly predictive, it may restrict the usefulness of the model in other operational contexts or rare disruption situations.

Next, even if the data leakage has been removed to an extent, some features can still be seen as proxies, obtaining signals that might correlate with the outcome.

Additionally, the dataset exhibits class imbalance, with fewer instances of high operational pressure events. While the model achieves strong recall for these events, precision is lower, indicating that some predicted high-pressure cases are false positives.

Another limitation of the classifier is that it does not include any features that indicate the temporal sequence of the incidents. To address this, a model that incorporates lag features from the preceding days was developed. This model achieved an ROC-AUC of 0.61, which indicates that while there is some predictive power in considering the temporal aspects of the incidents, there is also a need for more detailed data on the planned operational activities of LNER in the preceding days to increase the accuracy of the model.

Finally, evaluation focused primarily on accuracy and F1-score. Additional techniques such as ROC-AUC analysis and threshold tuning could further improve performance, particularly in balancing precision and recall for operational decision-making.

Forward-Looking Modelling Extension (MB W2066721)

Beyond the classification of incidents, a model was also developed that can help to forecast the incidents that may occur within any given day. The incident data was

aggregated to a daily level, and a threshold for high-pressure days was established (such as the 75th percentile of the total delay minutes for each day). Thus, the target variable for the model was defined established daily, rather than being inherited from the incident classifier.

Features used in the model included variables such as the total delay minutes for the previous day, the number of trains that were affected by incidents during the day, the number of train services that were cancelled due to those incidents, whether the previous day was recognized as a high-pressure day for the organization, and the average number of delay minutes for that train organization over periods of three or seven days. Furthermore, a train-test split was applied; the first 80% for training, while the remaining 20% is for testing the data

While the daily forecasting model achieved an ROC-AUC of 0.61 – an improvement over the baseline of 0.53, but not the level of performance required for operational deployment – it helps to indicate that there is no strong autoregressive nature to the operational pressure that is experienced on any given day. Despite having similar minutes of delay experienced on any given day, there can be significantly different causes for that delay, hence why the total number of delay minutes experienced on one day does not translate to a similar number of delay minutes the following day.

Instead, a more effective daily forecasting model would require into the operational environment would be one based on indicators of the daily operational load, such as maintenance load, fleet diagrams, possessions, and weather forecasts for the upcoming days. Furthermore, while these models are important for providing an estimation of the operational pressure that will be placed upon the organization on any given day, the model that has been developed and classified at the incident level scores a ROC-AUC of 0.91, indicating its value as the main output of this project.

Model Evaluation (MB W2066721)

While the model demonstrates strong recall (approximately 84%) for high operational pressure events, precision is lower (approximately 66%), indicating the presence of false positives. In the context of railway operations, false negatives are more critical than false positives, as failing to identify high-pressure events can lead to unexpected disruptions and significant financial costs.

Therefore, future improvements could involve adjusting the classification threshold to prioritise recall, even at the expense of precision, to better align with operational priorities.

Due to class imbalance within the dataset, accuracy alone is not a sufficient measure of performance. Therefore, additional metrics such as the F1 score were considered. The Random Forest model achieved a macro F1 score of approximately 0.82, providing a more balanced evaluation of performance across both classes. This confirms that the model performs well in identifying both normal and high-pressure conditions.

Although direct leakage variables were removed, the target variable is still derived from delay-based metrics, meaning the model partially relies on correlated features. Therefore, the model should be interpreted as detecting high-pressure conditions rather than fully predicting them in advance.

Operational Application During Disruption (MB W2066721)

The model could help LNER's Control and Engineering teams during live operations as an early warning. As an example, if the model classifies a particular depot or fleet class as high risk in the future, Control teams may be able to proactively, rather than reactively. They could locate spare rolling stock or move maintenance windows to avoid the peak. They could even swap diagrams before a shortage occurs. The timetable of LNER has been tightly optimised thanks to its 25% timetable enhancement. When working with very limited spare rolling stock, it only takes one unexpected fault to cause problems to cascade across the wider network. The model potentially provides teams with an opportunity to intervene rather than waiting for something to go wrong and then reacting. As noted during the client interview, cancellations carry a financial cost of over £10,000 per service. So, even occasional successful interventions would be of great operational and economic value. The model output could be integrated into existing Control room workflows as a daily risk dashboard that flags depots or periods that may need more scrutiny.

Forecast Horizon for Fleet Availability Risk (MB W2066721)

From the discussions in the client interview, a key operational consideration for LNER is how far ahead we should forecast. A forecast period of 24 to 48 hours is perhaps the most useful in practice. By this time, Control and Engineering teams have enough time to respond to a risk signal and the prediction does not become too uncertain to be relevant.

When the model flags a depot as high risk 24 hours in advance, Control and Engineering teams have sufficient time to respond. Subsequently, maintenance planning could adjust, spare rolling stock could be positioned, diagrams swapped in advance. A simple example to visualise, is a pushing / spare unit at a home depot with outstanding defects

that need maintenance work when away on service. Should we get 24-hour notice of a risk this allows sufficient time to pre-position this rolling stock along with a diagram swap. This effectively validates those practicalities of the 24-to-48-hour window.

Forecasts that extend beyond a couple of days can be beneficial for forward maintenance planning over several days or weeks. But as the railway is dynamic, such a prediction would become less accurate. Similarly, a prediction of simply a few hours does not offer any meaningful advantage.

Recommendations (MB W2066721)

- 1) **Prioritise maintenance intervention at high-incident depots** -The analysis at the depot level has identified the highest frequency of incidents of late start by IEP, BN, NL and EC. LNER should use these depots to formulate focused maintenance scheduling and resource allocation to reduce operational pressure at these locations.
- 2) **Establish a 24–48-hour operational risk monitoring system** - Our predictive models can identify periods of high operational pressure by examining patterns of historical delays. The Engineering and Control teams at LNER should develop rolling risk assessment practices. The risk assessment window should be between 24 and 48 hours. This will allow diagrams to be swapped proactively and spare rolling stock to be repositioned before the risk materializes into a live disruption.
- 3) **Investigate spike periods in delay frequency and severity** - Time series data analysis carried out which established spikes in delay frequency and delay severity. Notable spikes were identified across these operational time events, particularly in P2409 and P2410. More information must be captured for the seasons which experience an unusual level impact to identify the drivers behind them. Are these seasonal patterns, maintenance cycles or external factors? What is causing the unusually high operational impact? Once it is established, LNER will be on the front foot to manage these occurrences.
- 4) **Focus disruption interventions on highest-impact root causes** - Our root cause analysis shows that a small number of cause codes account for a disproportionately large share of average delay minutes. LNER must set out to target these repeat causes for the greatest saving in network delay minutes.
- 5) **Extend the framework using the developed daily lagged forecasting model**– A lagged forecasting model was built and tested on held out data, achieving an

ROC-AUC of 0.61 – an improvement on the baseline of 0.53. While this is a positive result, the model is still not one that can be reliably utilized in practice. The issue is that the model does not have visibility into what is planned for the LNER. Information from maintenance schedules, fleet diagrams, and engineering possessions would provide the model with some information on which to base its predictions. Making such information available to the model is one of the first steps that LNER should take.

An idea for future research is to use time-lagged features and make a prediction of “ahead-of-time” high operational pressure instead of classifying signals that relate to a disruption occurring at the same time.

Information about engineering and maintenance schedules could also be included in the prediction framework.

What are the main operational pain points caused by limited spare rolling stock? (MB W2066721)

The client interview and initial analysis make it clear that the LNER fleet has very limited capacity because of the 25% timetable growth. In other words, there are almost no additional trains beyond the numbers that will probably be used in the timetable. A single fault that takes place at a particular depot can therefore have a very rapid cascading effect. The cascading effect of every disruption will occur very quickly as there is no buffer of reserve trains available to cushion the pressure. The additional assessment calculates the existence of about 100 spare trains with the fleet over.

Which types of disruption cause the greatest systemic impact? (MB W2066721)

Based on the analysis, in terms of average delay minutes per incident, depot overhead line equipment faults (MU30), WSP electronics problems (MN21), and brake components defects (MN01) are all prominent causes of incident delays. According to the scatter plot (Figure 7), incidents that cause disruption to many trains bring about an outsized number of delay minutes. This confirms that multi-train disruption causes the systemic network pressure.

How is fleet availability currently assessed, and what are the limitations of existing processes? (MB W2066721)

According to the client interview, currently fleet availability management operation is largely reactive where operators do a manual check on maintenance and have access to different information about operations. Most of the available sensor data is not checked in an integrated manner nor are there any predictive checks. As a result, if there is any

risk to diagram coverage, this usually emerges too late to act. This closely fits within the problems space of the project brief.

Which visual indicators best represent availability risk? (MB W2066721)

These provide a nuanced view of where, when, and why availability pressure tends to manifest. According to the EDA, we note that the most operationally useful measures are delay incidents count by period (Figure 1), total delay minutes by period (Figure 2), late starts count at depot level (Figure 3), and root cause severity (Figure 8). The advised daily risk dashboard mentioned in the recommendation section incorporates these measures as its basis.

How could the developed model be utilised during disruption scenarios? (MB W2066721)

Two models were developed that can assist with decision making during operations. The model with an ROC-AUC of 0.91 can identify high operational pressure on the system at the point of the occurrence of incidents with that system. As such, the model can be incorporated into operational workflows to identify days where depots or fleet classes are likely to experience high operational pressure, allowing for Engineering and Control departments to begin to take steps to mitigate against a potential shortage of specific components. Additionally, the model with an ROC-AUC of 0.61 can provide a similar analysis but with a lead-time of 24-48 hours, utilizing features from the previous days to predict whether the next day is likely to have high operational pressure. Thus, each of these models can provide both real-time and forward-looking analysis to support a more proactive management of fleet availability in the system.

Conclusion (MB W2066721)

This project explored the use of machine learning models to analyse operational disruption data and identify periods of high operational pressure. While the initial models achieved very high accuracy, further analysis showed that this was partly due to strong relationships between input features and the target variable, meaning the models were not fully predictive.

To address this, the modelling approach was extended by introducing time-lagged features, allowing the model to use past operational data to predict future pressure

levels. Although this resulted in lower accuracy, it provides a more realistic representation of how such a system could be used in practice.

The results suggest that operational pressure is influenced by multiple interacting factors, including delay severity, incident frequency, and disruption patterns. While the current model provides a useful starting point, further improvements could be made by incorporating additional data sources such as weather conditions or infrastructure-related factors.

From a business perspective, even small improvements in prediction could have significant impact. For example, preventing just five cancellations per period, at an estimated cost of £10,000 per cancellation, could result in savings of £50,000 per period, or over £600,000 annually.

Substantial development has been made across the five project objectives. Through exploratory data analysis, the key risk factors have been identified. The highest risk fault types are depot overhead line faults, WSP electronics failures and brake component defects. Depot level analysis has identified Doncaster Carr, London Bounds Green, Leeds Neville Hill and Edinburgh Craiginny as the depots under the greatest operational pressure, which helps LNER establish clear maintenance priorities.

The Fleet Delay and Fleet Performance datasets demonstrated good data quality and completeness throughout the analysis. Nevertheless, the Fleet Planning Log required significant restructuring. In addition to this, the unavailability of engineering or fault-level data still poses a bottleneck for thorough analysis. Future iterations of this work would benefit from access to sensor-level or maintenance schedule data.

Github link (RRw2073057):

The complete project implementation, including all code and supporting materials, can be accessed via the GitHub repository linked below.

<https://github.com/Rayhanur798/LNER-Group-1/tree/main>

Referencing

(W1987563) DR

Dataset 1: Office of Rail and Road

Reason for selection:

This dataset records how many passenger trains are planned and key performance measures across operations in Great Britain. These measures include things such as, cancellations, significant lateness and public performance measure. The dataset gives us quarterly historical performance data for train operators. The information gathered provides us with reliable trends and service performance over time

Harvard reference

Office of Rail and Road, 2025. Table 3103 historic passenger trains planned, PPM and CaSL by operator. Available at: <https://dataportal.orr.gov.uk/statistics/performance/passenger-rail-performance/table-3103-historic-passenger-trains-planned-ppm-and-casl-by-operator/> Accessed 10 March 2026.

Dataset 2: Delay Minutes by Operator and Cause

Source: Office of Rail and Road

Reason for selection:

This dataset records the delays assigned to different causes for train operators in Great Britain. This includes causes such as infrastructure faults, signalling problems, external incidents, stock failures. This dataset supports the analysis of disruptions and helps find which factors contribute to the most delays across the network.

Harvard reference

Office of Rail and Road, 2025. Table 3184 delay minutes by operator and cause periodic. Available at: <https://dataportal.orr.gov.uk/statistics/performance/passenger-rail-performance/table-3184-delay-minutes-by-operator-and-cause-periodic/> Accessed 10 March 2026.

Dataset 3: NOAA Daily Weather Summaries Dataset

Source: National Oceanic and Atmospheric Administration, National Centres for Environmental Information

Reason for selection:

This dataset gives us the daily weather observations, which include such things as maximum temperature, minimum temperature, snowfall, wind speed and precipitation. Weather conditions influence railway operations and service reliability, so this dataset supports analysis of how environmental factors relate to train delays

Harvard reference

National Oceanic and Atmospheric Administration, 2025. Global Historical Climatology Network daily summaries dataset. Available at:

<https://www.ncei.noaa.gov/access/services/data/v1?dataset=daily-summaries&stations=UKE00107650&startDate=2018-01-01&endDate=2025-12-31&dataTypes=TMAX,TMIN,PRCP,SNOW,AWND&format=csv> Accessed 10 March 2026.

Office of Rail and Road (2025) Passenger rail performance statistics. Available at: <https://dataportal.orr.gov.uk/statistics/performance/passenger-rail-performance/> (Accessed: 10 March 2026).

Office of Rail and Road (2025) Table 3184 delay minutes by operator and cause periodic. Available at: <https://dataportal.orr.gov.uk/statistics/performance/passenger-rail-performance/table-3184-delay-minutes-by-operator-and-cause-periodic/> (Accessed: 10 March 2026).

Network Rail (2024) Rail performance and operations overview. Available at: <https://www.networkrail.co.uk> (Accessed: 10 March 2026).